SPIDER: A Faster and Smaller Identity-Based Signature From Isogenies

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Abstract. PRISM is an isogeny-based digital signature whose security relies on the problem of computing large prime degree isogenies from a supersingular elliptic curve of unknown endomorphism ring. In this paper, we design SPIDER, a digital signature that builds on a sigma protocol which when given a supersingular elliptic curve E, proves the knowledge of an isogeny $\phi : E \to E'$ of prime (or non smooth) degree d. SPIDER can hence be seen as a proof of knowledge of a PRISM signature. The latter observation allows us to combine SPIDER and PRISM to obtain an Identity-Based Signature (IBS) which we denote by SPIDER-IBS. SPIDER-IBS is the most efficient and the most compact isogeny-based IBS scheme, as all its predecessors are based on isogeny group actions.

Keywords: Post-Quantum Cryptography · Isogenies · identity-based Signature

1 Introduction

In the basic form of Public Key Cryptography (PKC), each user in the system generates their own key pair consisting of a public key and a corresponding secret key, and the user is assumed to be uniquely identified by their public key. In the real world however, users are generally not identified by randomly generated keys, but by more meaningful identities like their names or email addresses. To map public keys to real-world identities, a so-called public-key infrastructure (PKI) needs to be set up, involving a hierarchy of trusted certification authorities (CAs) that can certify public keys as belonging to a certain user. In this paradigm, the communication between two parties begins by the authentication of each other's identities, allowing them to make sure that their respective public keys are bound to their identities. To avoid this authentication step, Shamir [47] introduced Identity-Based Cryptography (IBC) by proposing Identity-Based Signatures (IBS) and Identity-Based Encryption (IBE) schemes where the public key of a user is their identity. In these protocols, the secret key of a user is issued by a trusted Key Generation Centre (KGC) who derives it from a master

secret that only it knows, and who is assumed to have an out-of-band method to verify the identity of the user. This eliminates some of the costs associated to PKIs and certificates. IBC is practical for closed systems such as army systems, enterprise systems, etc. where a user receives their secret key while integrating the system.

The IBS proposed by Shamir [47] is based on the integer factorization problem (RSA paradigm). Following this construction, many other IBS have been proposed using pairings on elliptic curves [31,54,45]. The security of these constructions is guaranteed by the presumed hardness of the integer factorization problem and the discrete logarithm problem. Hence as every scheme whose security relies on these problems, they are vulnerable to Shor's quantum attacks [50]. To address the quantum threat, there are several research directions in (post-quantum) cryptography, namely code-based cryptography, lattice-based cryptography, multivariate cryptography, isogeny-based cryptography, etc.

In the last decade, isogeny-based cryptography has gained a lot of interest because of the selection of SIKE [46] among the promising candidates for post-quantum key encapsulation schemes by NIST. The security of SIKE relied on the presumed hardness of computing an isogeny between two supersingular elliptic curves, given its degree, as well as the images of two points that form a basis of some accessible N torsion subgroup where N is smooth and larger than the degree (or its square root in some cases) of the said isogeny. Using the availability of this additional information (degree and torsion point images), three independent attacks [11,36,44] on SIDH [24] broke SIKE in polynomial time using isogenies between principally polarized abelian varieties of higher dimension (HD isogenies). Even though all isogeny-based schemes where this additional information was present (SIDH/SIKE type protocols) were broken by these attacks, they do not apply to schemes based on isogeny group actions (CSIDH [12] and friends [8,23,15]) or quaternions and isogenies (SQIsign [1] and friends). Most importantly, they offered a new way to represent isogenies. This new isogeny representation was leveraged in the design of many isogeny-based cryptosystems such as SQIsignHD [20] and the 2D variants [26,38,3] (that improve on SQISign [25]), (Q)FESTA [4,37], etc.

The first isogeny-based IBS has been proposed by Peng et al. [42]. This protocol is built using class group action and its security relies on that of CSI-FiSh [8]. In [49], Shaw and Dutta identified and addressed some shortcomings in the construction of Peng et al. [42]. In [14], Chen et al. proposed a tightly secure and more compact variant of this protocol. Due to the existence of sub-exponential quantum attacks [16] on isogeny group actions, several works [7,9,41,13] argue that using a 512 bits prime as the base field characteristic as originally proposed in CSIDH [12] does not offer enough quantum security in the isogeny group action setting. They [13] recommend using large prime characteristic of several thousand bits. This comes with a great efficiency and key/signature sizes overhead.

In [2], Basso et al. introduced PRISM, a new isogeny-based digital signature whose security relies on the problem of computing an isogeny of large

prime degree from an elliptic curve without knowing the endomorphism ring of the said curve. In PRISM, the signature is computed using the algorithm IdealToIsogeny [3], an algorithm allowing to compute the isogeny corresponding to a given quaternion ideal. Recently, Qlapoti [10] has been introduced as improvement of IdealToIsogeny, it leads to a more efficient instantiation to PRISM.

Contribution. Our first contribution is a proof of knowledge (POK) of an isogeny of given degree. In fact, existing proofs of an isogeny of given degree assume that the codomain of this isogeny is known, that is the relation at play is

$$\mathcal{R}_{isog-d} = \{((E_0, E), \phi),\ \phi : E_0 \to E \text{ and } \deg \phi = d\}.$$

In this paper, we consider the setting where the codomain of the isogeny is not part of the statement in the relation. More precisely, we design a POK for the relation

$$\mathcal{R}_d = \{(E, \phi) \mid \exists E' \text{ such that } \phi : E \to E' \text{ and } \deg \phi = d\}.$$

The relation $\mathcal{R}_d$ is not hard when d is a smooth integer, as it is easy to generate smooth degree isogenies from any given supersingular elliptic curve E. Nevertheless, when d is prime or has a large prime factor, $\mathcal{R}_d$ is believed to be a hard relation. In this setting, we turn our POK into a secure digital signature scheme, SPIDER.

Our second contribution is to combine SPIDER and PRISM to obtain an IBS scheme. The hardness of the relation $\mathcal{R}_d$ for degrees $d = q(2^a - q)$ where q is a large prime underlies the security of the PRISM signature [2]. In PRISM, a signature is in fact such an isogeny, a witness in the relation $\mathcal{R}_d$ with $d = q(2^a - q)$. This implies that SPIDER is a POK of a PRISM signature. One can hence design an IBS as follows. The master secret/public key pair is a PRISM secret/public key pair. To generate the secret key corresponding to a user id id, the master whose public key is a supersingular elliptic curve E signs the id using PRISM to obtain a PRISM signature $\phi_{\mathsf{id}} : E \to E'$ which is an isogeny of degree $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$ where q_{id} is a large prime tied to the identity id (in practice, q_{id} is obtained by hashing E and id). Also, $\phi_{\mathsf{id}} : E \to E'$ is a witness for E in the relation $\mathcal{R}_{d_{\mathsf{id}}}$. They return $\phi_{\mathsf{id}} : E \to E'$ to the user. Since $\phi_{\mathsf{id}} : E \to E'$ is a valid SPIDER secret key, the user uses it to sign (using the SPIDER signature scheme this time) messages. Given a signature σ returned by a user whose id is id, the verifier checks that the SPIDER signature is valid with respect to $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$. We name this identity-based signature scheme SPIDER-IBS. To prove the security of SPIDER-IBS, we introduce a new assumption. This assumption, at a high level, states that the SPIDER signature scheme with respect to the relation $\mathcal{R}_d$ where $d = q(2^a - q)$ (with q being a large prime) remains secure when the adversary is given access to a special degree isogeny oracle (SPEDIO) which they can query to get isogenies of degree $q'(2^a - q')$ where $q' \neq q$.

Lastly, we provide parameters for our IBS and compare it to existing isogeny-based IBS schemes. To the best of our knowledge, all existing [48,14] isogeny-based IBS schemes prior to this work are built in the supersingular isogeny group action framework. This implies that they need to use a large prime characteristic to ensure enough quantum security [13]. This is not the case for SPIDER-IBS, as it can be instantiated with much smaller primes. In fact, for the NIST level I security level, we insatiate our scheme with a prime of about 512 bits. Compared to the isogeny group action based IBS schemes in [48,14] instantiated with a 3000 bits prime, for similar signature sizes, the master public key and the user secret key of the isogeny group action based IBS scheme are about 14.42x and 30.97x larger respectively. When it comes to efficiency, SPIDER-IBS is several orders of magnitude more efficient. For example, in the isogeny group action based IBS scheme, the master key generation, the user key generation, the signature and the verification cost about 192 000, 186 000, 156 000 and 156 000 (2,2,2,2) isogenies in characteristic p with $\log_2 p \approx 3000$. Meanwhile in SPIDER-IBS, the master key generation and the user key generation cost about 512 $(2,2)$ isogenies over $\mathbb{F}_{p^2}$ each, with $\log_2 p \approx 512$; the signing costs about 243 dimension 1 isogenies of degree $2^{2\lambda}$, the verification costs 177 dimension 1 isogenies of degree $2^{2\lambda}$ and 8448 (2,2) isogenies over $\mathbb{F}_{p^2}$ each.

Outline. The remainder of the paper is organized as follows. In Section 2 we recall the necessary background needed to understand this work. In Section 3 we describe our new POK for the relation $\mathcal{R}_d$. Section 4 is dedicated to SPIDER. In Section 5, we describe SPIDER-IBS and its security arguments, provide parameters for SPIDER and SPIDER-IBS and compare SPIDER-IBS to existing isogeny-based IBS schemes.

2 Preliminaries

2.1 Sigma Protocols and digital signatures

The identification scheme we propose in this paper is constructed in the sigma protocol paradigm.

Definition 1 (Sigma Protocol). *Let $\mathcal{R}$ be a relation. A sigma protocol Σ for $\mathcal{R}$ is a three-move proof system that consists of (Probabilistic Polynomial Time) PPT algorithms $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2)$ and $\mathbf{V} = (\mathbf{V}_1, \mathbf{V}_2)$, where $\mathbf{V}_2$ is deterministic. The algorithms $\mathbf{P}_1$ and $\mathbf{P}_2$ share their internal states, as do $\mathbf{V}_1$ and $\mathbf{V}_2$. Let* ChSet *denote a challenge space. Then, Σ proceeds as follows.*

- *The prover, on input* $(\mathsf{stm}, \mathsf{w}) \in \mathcal{R}$*, runs* $(\mathsf{com}) \leftarrow \mathbf{P}_1(\mathsf{stm}, \mathsf{w})$ *and sends a* commitment com *to the verifier.*
- *The verifier runs* $\mathsf{ch} \leftarrow \mathbf{V}_1(\mathsf{stm}, \mathsf{com})$*, drawing a uniformly random challenge from* ChSet*, and sends it to the prover.*
- *The prover, given* ch*, runs* $\mathsf{resp} \leftarrow \mathbf{P}_2(\mathsf{stm}, \mathsf{w}, \mathsf{com}, \mathsf{ch})$ *and returns a* response resp *to the verifier.*

– *The verifier runs* $\mathbf{V}_2(\mathsf{stm}, \mathsf{com}, \mathsf{ch}, \mathsf{resp})$ *and outputs* $\mathsf{accepted}$ *or* $\mathsf{rejected}$

The protocol transcript $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$ *is said to be valid when* $\mathbf{V}_2(\mathsf{stm}, \mathsf{com}, \mathsf{ch}, \mathsf{resp})$ *outputs* $\mathsf{accepted}$.

We require the sigma protocol to be correct:

Definition 2 (Correctness). *A sigma protocol* Σ *is said to be correct if for all* $(\mathsf{stm}, \mathsf{w}) \in \mathcal{R}$,

$$\Pr\left[\mathbf{V}_2(\mathsf{stm}, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = \mathsf{accepted} \;\middle|\; \begin{array}{c} \mathsf{com} \leftarrow \mathbf{P}_1(\mathsf{stm}, \mathsf{w}), \\ \mathsf{ch} \leftarrow \mathbf{V}_1(\mathsf{stm}, \mathsf{com}), \\ \mathsf{resp} \leftarrow \mathbf{P}_2(\mathsf{stm}, \mathsf{w}, \mathsf{com}, \mathsf{ch}) \end{array}\right] = 1.$$

Once a sigma protocol is correct, the security properties are *Honest Verifier Zero-Knowledge* and *special soundness*.

Definition 3 (Honest Verifier Zero-Knowledge). *We say* Σ *is* honest-verifier-zero-knowledge*(HVZK) for a relation* $\mathcal{R}$ *if there exists a PPT simulator* **Sim** *such that for any statement-witness pair* $(\mathsf{stm}, \mathsf{w}) \in \mathcal{R}$, $\mathsf{ch} \in \mathsf{ChSet}$, *any computationally unbounded*[3] *adversary* $\mathcal{A}$ *and any security parameter* λ *we have*

$$\left| \begin{array}{l} \Pr\left[\mathcal{A}(\mathsf{com}, \mathsf{ch}, \mathsf{resp}) = \mathsf{accepted} \;\middle|\; \begin{array}{c} \mathsf{com} \leftarrow \mathbf{P}_1(\mathsf{stm}, \mathsf{w}) \\ \mathsf{resp} \leftarrow \mathbf{P}_2(\mathsf{stm}, \mathsf{w}, \mathsf{ch}) \end{array}\right] \\ \\ -\Pr\left[\mathcal{A}(\mathsf{com}, \mathsf{ch}, \mathsf{resp}) = \mathsf{accepted} \;\middle|\; (\mathsf{com}, \mathsf{resp}) \leftarrow \mathbf{Sim}(\mathsf{stm}, \mathsf{ch})\right] \end{array} \right| = \mathsf{negl}(\lambda)$$

Definition 4 (Special Soundness). *We say* Σ *has* 2*-special soundness for the relation* $\mathcal{R}$ *if there exists a polynomial-time extraction algorithm* **Extract** *such that, given a statement* stm *and any* 2 *valid transcripts* $(\mathsf{com}, \mathsf{ch}_1, \mathsf{resp}_1), (\mathsf{com}, \mathsf{ch}_2, \mathsf{resp}_2)$ *relative to* stm *with distinct challenges* $\mathsf{ch}_1, \mathsf{ch}_2$ *and the same commitment* com, **Extract** *outputs a witness* w *satisfying* $(\mathsf{stm}, \mathsf{w}) \in \mathcal{R}$.

Definition 5. *A digital signature scheme consists of four probabilistic polynomial time algorithms* **SetUp**, **KeyGen**, **Sign** *and* **Verify** *designed as follows:*

- $\mathbf{SetUp}(1^\lambda)$*: Takes the security parameter* λ *as input and returns the public parameters* pp.
- $\mathbf{KeyGen}(\mathsf{pp})$*: Takes the public parameters* pp *as input and returns a pair* $(\mathsf{pk}, \mathsf{sk})$ *of public and secret keys respectively.*
- $\mathbf{Sign}(\mathsf{pp}, \mathsf{sk}, \mathsf{pk}, \mathsf{msg})$*: Takes the public parameters* pp*, the public key* pk *and the associated secret key* sk*, and a message* msg *as inputs, and returns a signature* σ *for the message* msg.
- $\mathbf{Verify}(\mathsf{pp}, \mathsf{pk}, \mathsf{msg}, \sigma)$*: Returns* $\mathsf{accepted}$ *or* $\mathsf{rejected}$ *indicating whether the signature verification was successful or not.*

[3] This precision corresponds to the statistical HVZK. When the adversary is computationally bounded, it is the computational HVZK.

Definition 6 (Correctness). *A digital signature scheme* $(\mathbf{SetUp}, \mathbf{KeyGen}, \mathbf{Sign}, \mathbf{Verify})$ *is said to be correct if*

$$\Pr\left[\mathbf{Verify}(\mathsf{pp}, \mathsf{pk}, \mathsf{msg}, \sigma) = \mathsf{accepted} \;\middle|\; \begin{array}{c} \mathsf{pp} \leftarrow \mathbf{SetUp}(1^\lambda), \\ (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathbf{KeyGen}(\mathsf{pp}), \\ \sigma \leftarrow \mathbf{Sign}(\mathsf{pp}, \mathsf{sk}, \mathsf{pk}, \mathsf{msg}) \end{array}\right] = 1.$$

Definition 7 (EUF-CMA: Existential UnForgeability against Chosen-Message Attack). *A digital signature scheme* $(\mathbf{SetUp}, \mathbf{KeyGen}, \mathbf{Sign}, \mathbf{Verify})$ *is said to be EUF-CMA secure if any adversary* $\mathcal{A}$ *has a negligible advantage in the following game against a challenger* $\mathcal{C}$*, where* λ *is the security parameter, the advantage being the probability of winning the game.*

EUF-CMA game:
- $\mathcal{C}$ *runs* **SetUp** *on input* 1^λ *to obtain public parameters* pp*, then runs* **KeyGen** *on input* pp *to obtain a key pair* $(\mathsf{pk}, \mathsf{sk})$*. It gives* pp *and* pk *to* $\mathcal{A}$ *and keeps* sk *secret. They initiate an empty list* S_{msg} *of signed messages.*
- $\mathcal{A}$ *makes polynomially many adaptive queries to the signing oracle* O_{sign} *and outputs a pair* $(\mathsf{msg}_*, \sigma_*)$*.*
- $\mathcal{C}$ *accepts* $(\mathsf{msg}_*, \sigma_*)$ *if* $\mathsf{msg}_* \notin S_{\mathsf{msg}}$ *and* $\mathbf{Verify}(\mathsf{pp}, \mathsf{pk}, \mathsf{msg}_*, \sigma_*) = \mathsf{accepted}$*.*

Oracle $O_{sign}(\mathsf{msg})$*:*
- $\sigma \leftarrow \mathbf{Sign}(\mathsf{pp}, \mathsf{sk}, \mathsf{pk}, \mathsf{msg})$
- $S_{\mathsf{msg}} \leftarrow S_{\mathsf{msg}} \cup \{\mathsf{msg}\}$

We say that $\mathcal{A}$ *wins if* $\mathcal{C}$ *accepts* $(\mathsf{msg}_*, \sigma_*)$*.*

Whenever a sigma protocol satisfies the above requirements (Correctness, HVZK an special soundness), it can be turned into signature scheme thanks to Fiat-Shamir transform [27]. The obtained signature scheme is EUF-CMA secure and is constructed as follows, where $H : \{0,1\}^* \to \mathsf{ChSet}$ is a hash function, modelled as random oracle:

$\mathbf{SetUp}(1^\lambda)$:
- Return the public parameters pp

$\mathbf{KeyGen}(\mathsf{pp})$:
- Return $(\mathsf{pk}, \mathsf{sk})$

$\mathbf{Sign}(\mathsf{pp}, \mathsf{sk}, \mathsf{pk}, \mathsf{msg})$:
- $\mathsf{com} \leftarrow \mathbf{P}_1(\mathsf{pk}, \mathsf{sk})$
- $\mathsf{ch} \leftarrow H(\mathsf{pk}||\mathsf{com}||\mathsf{msg})$
- $\mathsf{resp} \leftarrow \mathbf{P}_2(\mathsf{pk}, \mathsf{sk}, \mathsf{com}, \mathsf{ch})$
- Return $\sigma = (\mathsf{com}, \mathsf{resp})$

$\mathbf{Verify}(\mathsf{pk}, \mathsf{msg}, \sigma)$:
- $(\mathsf{com}, \mathsf{resp}) \leftarrow \sigma$
- $\mathsf{ch} \leftarrow H(\mathsf{pk}||\mathsf{com}||\mathsf{msg})$
- Return $\mathbf{V}_2(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$

Fig. 1. Fiat-Shamir Transform

2.2 Identity-Based Signature

Definition 8. *An identity-based signature scheme (IBS) consists of five PPT algorithms* **SetUp**, **MKeyGen**, **KeyDer**, **Sign** *and* **Verify** *designed as follows:*

- $\mathbf{SetUp}(1^\lambda)$*: Takes the security parameter* λ *as input and returns the public parameters* pp.
- $\mathbf{MKeyGen}(\mathsf{pp})$ *Takes the public parameters* pp *as input and outputs the master public key* mpk *and the master secret key* msk.
- $\mathbf{KeyDer}(\mathsf{pp}, \mathsf{msk}, \mathsf{mpk}, \mathsf{id})$*: Takes the public parameters* pp*, the master secret key* msk*, the master public key* mpk *and a user identity* id *as inputs, and returns a user secret key* usk.
- $\mathbf{Sign}(\mathsf{pp}, \mathsf{mpk}, \mathsf{usk}, \mathsf{id}, \mathsf{msg})$*: Takes the public parameters* pp*, the master public key* mpk*, a user secret key* usk*, a user identity* id *and a message* msg *as input, and returns a signature* σ *for message* msg.
- $\mathbf{Verify}(\mathsf{pp}, \mathsf{mpk}, \mathsf{id}, \mathsf{msg}, \sigma)$*: Return* $\mathsf{accepted}$ *or* $\mathsf{rejected}$ *indicating whether signature verification was successful or not.*

Definition 9 (Correctness). *An identity-based signature scheme* $\mathcal{IBS} = (\mathbf{SetUp}, \mathbf{MKeyGen}, \mathbf{KeyDer}, \mathbf{Sign}, \mathbf{Verify})$ *is said to be correct if for all identity* id

$$\Pr\left[\mathbf{Verify}(\mathsf{mpk}, \mathsf{id}, \mathsf{msg}, \sigma) = \mathsf{accepted} \;\middle|\; \begin{array}{c} \mathsf{pp} \leftarrow \mathbf{SetUp}(1^\lambda), \\ \mathsf{mpk}, \mathsf{msk} \leftarrow \mathbf{MKeyGen}(\mathsf{pp}) \\ \mathsf{usk} \leftarrow \mathbf{KeyDer}(\mathsf{pp}, \mathsf{msk}, \mathsf{id}), \\ \sigma \leftarrow \mathbf{Sign}(\mathsf{mpk}, \mathsf{usk}, \mathsf{id}, \mathsf{msg}) \end{array}\right] = 1.$$

Definition 10 (UnForgeability against Chosen-Identity and Chosen Message Attacks). *Let* $\mathcal{IBS} = (\mathbf{SetUp}, \mathbf{MKeyGen}, \mathbf{KeyDer}, \mathbf{Sign}, \mathbf{Verify})$ *be an IBS,* $\mathcal{A}$ *an adversary and* λ *a security parameter. The* UF-CID-CMA advantage *of* $\mathcal{A}$ *is its success probability*

$$\mathbf{Adv}^{\mathcal{IBS}}_{\mathcal{A}}(\lambda) = \Pr[\mathsf{Exp}^{IBS}_{\lambda}(\mathcal{A}) = 1]$$

in the experiment $\mathsf{Exp}^{IBS}_{\lambda}(\cdot)$ *(Figure 2). We say that* $\mathcal{IBS}$ *is* UF-CID-CMA secure *if* $\mathbf{Adv}^{\mathcal{IBS}}_{\mathcal{A}}(.)$ *is negligible for every polynomial time adversary* $\mathcal{A}$.

Input: The challenger $\mathcal{C}$ takes as input the security parameter λ, and generates $\mathsf{pp} \leftarrow \mathbf{SetUp}(1^\lambda)$ and $(\mathsf{mpk}, \mathsf{msk}) \leftarrow \mathbf{MKeyGen}(\mathsf{pp})$. It gives the public parameters pp and the master public key mpk to the adversary $\mathcal{A}$ and keeps the master secret key msk. They initiate empty lists C_{id} and S_{msg} of corrupted identities and signed messages.

Query Phase: $\mathcal{C}$ responds to polynomially many adaptive queries made by $\mathcal{A}$,

- Oracle $O_{\mathsf{corr}}()$: On receiving queries on a user identity id from $\mathcal{A}$, the challenger responds with the corresponding user secret key $\mathsf{usk} \leftarrow \mathbf{KeyDer}(\mathsf{pp}, \mathsf{msk}, \mathsf{id})$ and adds id to the list C_{id}.
- Oracle $O_{\mathsf{sign}}()$: On receiving queries on a message msg, and a user identity id from the adversary $\mathcal{A}$, the challenger responds with a signature $\sigma \leftarrow \mathbf{Sign}(\mathsf{mpk}, \mathsf{usk}, \mathsf{id}, \mathsf{msg})$ where $\mathsf{usk} \leftarrow \mathbf{KeyDer}(\mathsf{pp}, \mathsf{msk}, \mathsf{id})$ is the user secret key corresponding to the identity id. The challenger adds $(\mathsf{msg}, \mathsf{id})$ to the list S_{msg}.

Forgery: $\mathcal{A}$ eventually outputs a message msg^*, a user identity id^*, and a forgery σ^*. $\mathcal{A}$ wins the game if $\mathbf{Verify}(\mathsf{mpk}, \mathsf{id}^*, \mathsf{msg}^*, \sigma^*) = \mathsf{accepted}$, $\mathsf{id}^* \notin C_{\mathsf{id}}$ and $(\mathsf{msg}^*, \mathsf{id}^*) \notin S_{\mathsf{msg}}$.

Fig. 2. Security experiment $\mathsf{Exp}_\lambda^{\mathrm{IBS}}(\mathcal{A})$

2.3 Background on Isogenies

In this section, we recall some preliminaries on elliptic curves and isogenies that can help to understand this work. For further details on these topics, we recommend De Feo notes [22] or Silverman's book [51, Chap. 3,5].

Let E and E' be two elliptic curves. An *isogeny* of domain E and codomain E' is a non zero rational map $\phi : E \to E'$ that is also a group homomorphism. The degree of an isogeny is its degree as rational map. One says that an isogeny is separable if its degree is prime to p. In this case, its degree corresponds to the cardinality of its kernel and the isogeny is uniquely determined (up to isomorphism of its codomain) by its kernel. In this work, all our isogenies are separable. Given an isogeny $\phi : E \to E'$, there exists a unique isogeny $\widehat{\phi} : E' \to E$ verifying $\widehat{\phi} \circ \phi = [\deg \phi]_E$ and $\phi \circ \widehat{\phi} = [\deg \phi]_{E'}$, where $[\deg \phi]_E$ and $[\deg \phi]_{E'}$ are the multiplication by $\deg \phi$ (the degree of ϕ) in E and E' respectively. The isogeny $\widehat{\phi}$ is called *the dual isogeny of* ϕ. Let $\phi : E \to E_1$ and $\psi : E \to E_2$ be two separable isogenies of coprime degree. The *pushforward of* ψ *through* ϕ is the isogeny $\psi' = [\phi]_*\psi$ of kernel $\ker([\phi]_*\psi) = \phi(\ker \psi)$. Having this, ψ is called *pullback* of $[\phi]_*\psi$ through ϕ. One notes this by $\psi = [\phi]^*\psi'$.

An isogeny from E to itself is called endomorphism of E. Together with the zero map, these endomorphisms form a ring where the addition is derived from the group low and the multiplication is the composition. This ring is called the endomorphism ring of E and is denoted by $\mathrm{End}(E)$. The problem of computing

the endomorphism ring of a random supersingular elliptic curve is believed to be hard, and underlies the security of isogeny-based cryptography.

Prior to the attacks on SIDH [11,36,44], the only efficient representation of an isogeny when the endomorphism ring of the domain curve is not known was the kernel representation, which consists of representing the isogeny by a generator of its kernel. Using this representation, the isogeny is computed thanks to Vélu's formulas [52] or $\sqrt{\text{élu}}$ [6]. This approach is limited to compositions of rational isogenies of smooth degree. For example, it cannot be used for isogenies of large prime degree. The attacks on SIDH can be seen as an interpolation [43] of an isogeny once we have its degree and its action on some accessible torsion points of large smooth order. They hence lead to a new isogeny representation where an isogeny $\phi : E \to E'$ of degree d is represented by $\phi(E[N])$ where $N > d$ is smooth and the N torsion is accessible. The main tool of these attacks is the following theorem from Kani [32]. The following formulation is taken from [2].

Theorem 11. *Let d_1 and d_2 be coprime integers such that $N = d_1 + d_2$ is coprime with p, and let E_0, E_1, E_2, and E_3 be elliptic curves connected by the following diagram of isogenies:*

$$\begin{array}{ccc} E_0 & \xrightarrow{\varphi_1} & E_1 \\ {\scriptstyle\psi_2}\downarrow & & \downarrow{\scriptstyle\varphi_2} \\ E_2 & \xrightarrow[\psi_1]{} & E_3 \end{array}$$

such that $\deg(\varphi_1) = \deg(\psi_1) = d_1$, $\deg(\varphi_2) = \deg(\psi_2) = d_2$ *and* $\varphi_2 \circ \varphi_1 = \psi_1 \circ \psi_2$. *Then the map*

$$\Phi = \begin{pmatrix} \varphi_1 & \widehat{\varphi_2} \\ -\psi_2 & \widehat{\psi_1} \end{pmatrix} : E_0 \times E_3 \longrightarrow E_1 \times E_2$$

is an isogeny of (principally polarized) abelian varieties with kernel

$$\ker(\Phi) = \left\{ \left(\widehat{\varphi_1}(P), \varphi_2(P)\right) \;\middle|\; P \in E_1[N] \right\} \;\cong\; \tfrac{\mathbb{Z}}{N\mathbb{Z}} \times \tfrac{\mathbb{Z}}{N\mathbb{Z}}.$$

Assuming that N is powersmooth, or N is smooth and all N-torsion points are rational, the isogeny Φ can be efficiently evaluated at any point on $E_0 \times E_3$. For example, if $N = 2^a$ for some $a \geq 1$ then one can use theta coordinates [21,35] to compute Φ. In this case, the generators of the kernel defining Φ encode an efficient two-dimensional representation of φ_1.

However, this does not solve the problem of computing an isogeny of large prime degree from a random supersingular elliptic curve. In fact, the previous theorem applies once we have an evaluation of such an isogeny on N-torsion points, and having such evaluation is not obvious. We summarize this task in the following Problem.

Problem 12. Let E be a random supersingular elliptic curve and q a large prime. Compute an isogeny of degree q and domain E.

When the endomorphism ring of the starting curve E is known, this problem can be solved efficiently [3,10] thanks to the Deuring correspondence. For a random curve, the best known algorithm has complexity $\tilde{O}(q^2)$ [2]. This problem underlies the security of PRISM [2].

2.4 Quaternion Algebras and Deuring correspondence

In this subsection, we give some preliminary notions in quaternion algebras and Deuring correspondence. For more details, we recommend the thesis of Antonin Leroux [34, Chap. 1,2,3] or John Voight's book [53, Chap. 10,16,17]

A *quaternion algebras* over $\mathbb{Q}$ is an algebra $\left(\frac{a,b}{\mathbb{Q}}\right)$ of dimension 4 generated over $\mathbb{Q}$ by $1, i, j, k$ which satisfy $i^2 = a$, $j^2 = b$ and $k = ij = -ji$, for some $a, b \in \mathbb{Z}$.

Let $x = x_1 + x_2 i + x_3 j + x_4 k \in \left(\frac{a,b}{\mathbb{Q}}\right)$. The conjugate of x is $\bar{x} = x_1 - (x_2 i + x_3 j + x_4 k)$, its norm is $n(x) = x\bar{x}$ and its trace is $tr(x) = x + \bar{x}$.

An order in $\left(\frac{a,b}{\mathbb{Q}}\right)$ is a $\mathbb{Z}$-submodule $\mathcal{O}$ of dimension 4 that is also a subring. It is maximal if it is not contained in another order. Given two maximal orders $\mathcal{O}_1$ and $\mathcal{O}_2$, there exists a left ideal $I(\mathcal{O}_1, \mathcal{O}_2)$ of $\mathcal{O}_1$ that is a right ideal of $\mathcal{O}_2$. $I(\mathcal{O}_1, \mathcal{O}_2)$ is called connecting ideal of $\mathcal{O}_1$ and $\mathcal{O}_2$.

Given a place v in $\mathbb{Q}$ (v is a prime or ∞), one says that $\left(\frac{a,b}{\mathbb{Q}}\right)$ is ramified at v if $\left(\frac{a,b}{\mathbb{Q}}\right) \otimes \mathbb{Q}_v$ is not isomorphic to the algebra of 2×2 matrix over $\mathbb{Q}_v$, where $\mathbb{Q}_v$ is $\mathbb{R}$ if $v = \infty$ or the field of v-adic numbers if v is prime. The places of ramification uniquely determine a quaternion algebras up to isomorphism. We are interested in the quaternion algebras ramified at p and ∞, that one notes $\mathcal{B}_{p,\infty}$. For instance, when $p \equiv 3 \bmod 4$, we have $\mathcal{B}_{p,\infty} = \left(\frac{-1,-p}{\mathbb{Q}}\right)$.

There exists a correspondence between the isomorphism classes of supersingular elliptic curve (up to Galois conjugacy) and maximal orders in $\mathcal{B}_{p,\infty}$. This correspondence is called *Deuring correspondence*.

Example 13. For $p \equiv 3 \bmod 4$, the elliptic curve $E_0 : y^2 = x^3 + x$ is supersingular. Its endomorphism ring is isomorphic to $\mathcal{O}_0 = \mathbb{Z}\langle 1, i, \frac{i+j}{2}, \frac{1+k}{2}\rangle$ through the isomorphism $i \mapsto \iota$, $j \mapsto \pi$, where $\iota : (x, y) \mapsto (-x, \sqrt{-1}y)$ and $\pi : (x, y) \mapsto (x^p, y^p)$.

The following Table 1 summarises this correspondence.

Given a supersingular elliptic curve E and a left ideal of $\mathcal{O} \cong \mathrm{End}(E)$, the corresponding isogeny can be computed using Qlapoti [10]

2.5 PRISM

PRISM [2] is a hash-and-sign like digital signature whose security relies on the problem of computing an isogeny of large prime degree without knowing the endomorphism ring of the domain curve. It relies on the following algorithms:

Table 1. A summary of the Deuring correspondence

Supersingular j-invariant over $\mathbb{F}_{p^2}$	**Quaternion object in $\mathcal{B}_{p,\infty}$**
$j(E)$ (up to Galois conjugacy)	$\mathcal{O} \cong \mathrm{End}(E)$ (up to isomorphism)
(E_1, φ) with $\varphi : E \to E_1$	I_φ integral left $\mathcal{O}$-ideal and right $\mathcal{O}_1$-ideal
$\theta \in \mathrm{End}(E)$	Principal ideal $\mathcal{O}\theta$
$\deg(\varphi)$	$n(I_\varphi)$ (the reduced norm)
$\widehat{\varphi}$	$\overline{I_\varphi}$ (the conjugate ideal)
$\varphi : E \to E_1$, $\psi : E \to E_1$	Equivalent ideals $I_\varphi \sim I_\psi$
$\tau \circ \rho : E \to E_1 \to E_2$	$I_{\tau\circ\rho} = I_\rho \cdot I_\tau$

- $\mathsf{GenIsogeny}(E, \phi, q)$: On input a supersingular elliptic curve E, an isogeny $\phi : E_0 \to E$ and a prime $q \in (2^{a-1}, 2^a)$ for some integer a, it returns an efficient representation of an isogeny $\varphi : E \to E'$ of degree $q(2^a - q)$. At a high level,that is done by sampling a uniformly random ideal of norm $q(2^a - q)$ and computing the corresponding isogeny using the ideal to isogeny algorithm $\mathsf{Qlapoti}$ [10].
- $\mathsf{VerIsogeny}(\varphi, E, q)$: On input a supersingular elliptic curve E, an isogeny $\varphi : E \to E'$ and a prime $q \in (2^{a-1}, 2^a)$, it verifies that $\deg(\varphi) = q(2^a - q)$.

We now describe the PRISM signature algorithm.

Public parameters. Set p, a prime congruent to 3 modulo 4. Set $E_0/\mathbb{F}_{p^2}$, a supersingular elliptic curve of known endomorphism ring. Set a, a positive integer such that $E_0[2^a] \subset E_0(\mathbb{F}_{p^2})$. Let $\mathsf{primes_a} = \{q,\ \text{prime s.t } 2^{a-1} < q < 2^a\}$ be the set of a bits primes. Set $\mathsf{H}_{\mathsf{primes}_a} : \{0,1\}^* \to \mathsf{primes_a}$ a hash function. We have $\mathsf{pp}_{\mathsf{PRISM}} = \{p, E_0, a, \mathsf{H}_{\mathsf{primes}_a}\}$.

Key Generation. Compute a uniformly random isogeny $\phi_{\mathsf{sk}} : E_0 \to E_{\mathsf{pk}}$ and set $\mathsf{sk} = \phi_{\mathsf{sk}}$ and $\mathsf{pk} = E_{\mathsf{pk}}$. The isogeny ϕ_{sk} is sampled in such a way that the distribution of E_{pk} negligibly close to the uniform distribution in the supersingular isogeny graph. For example, it suffices to sample ϕ_{sk} as a random isogeny of prime degree $d_0 > p \cdot 2^{2\lambda}$.

Signing. To sign a message msg, one computes $\mathsf{H}_{\mathsf{primes}_a}(E_{\mathsf{pk}}||\mathsf{msg})$ to obtain a prime q having a bits. The signature σ is a representation of a uniformly random isogeny $\varphi : E \to E'$ of degree $q(2^a - q)$, computed using $\mathsf{GenIsogeny}$.

Verification. To verify a signature σ on a message msg, one computes $q = \mathsf{H}_{\mathsf{primes}_a}(E_{\mathsf{pk}}||\mathsf{msg})$, then one calls $\mathsf{VerIsogeny}$ on σ, E_{pk} and q to verify that σ is a valid representation of an isogeny $\varphi : E_{\mathsf{pk}} \to E'$ of degree $q(2^a - q)$.

Each PRISM signature provides a $q(2^a - q)$-isogeny which is hard to generate in practice. In the security analysis of PRISM, an oracle generating such isogenies

is provided to the adversary. This oracle is called SPEDIO. The signature scheme is then proven [2, Prop. 2] to be EUF-CMA secure under the assumption that $\mathsf{H}_{\mathsf{primes}_a}$ is a collision resistant hash function and Problem 15 is hard.

Definition 14. *A special degree isogeny oracle (SPEDIO) is an oracle which takes as inputs a supersingular elliptic curve E defined over $\mathbb{F}_{p^2}$ and a prime $q \in \mathsf{primes_a}$, and returns a uniformly random cyclic isogeny of degree $q(2^a - q)$ from E.*

Problem 15. Given a random supersingular elliptic curve E and a SPEDIO, output an isogeny of degree $q'(2^a - q')$ with $q' \in \mathsf{primes_a}$ different from all degrees q formerly generated by the oracle.

Proposition 16 ([2, Prop. 2]). *If $\mathsf{H}_{\mathsf{primes}_a}$ is a collision-resistant cryptographic hash function and Problem 15 is hard, then PRISM is EUF-CMA secure.*

Now we focus on proving the knowledge of some isogeny of large prime/non smooth degree from a given curve E. This POK is the main tool for the construction of our IBS.

3 Proof of knowledge of a prime degree isogeny

In this section, we design a sigma protocol that allows a prover to prove the knowledge of an isogeny of degree d and domain E. This proof of knowledge will later be used to prove the knowledge of PRISM signatures, and hence play a crucial role in the design of our IBS.

We consider the following relation $\mathcal{R}_d$

$$\mathcal{R}_d = \{(E, \phi),\ |\exists E' \text{ such that } \phi : E \to E' \text{ and } \deg \phi = d\}$$

where the statements are supersingular elliptic curves E and a witness for E is a degree d isogeny $\phi : E \to E'$. There exists a SIDH proof of knowledge for the case where the curve E' is part of the statement and the witness is made of the isogeny ϕ only. The latter proof is a proof of knowledge for the following relation

$$\mathcal{R}_{isog-d} = \{((E_0, E), \phi),\ \phi : E_0 \to E \text{ and } \deg \phi = d\}.$$

The SIDH proof relies on the following commutative diagram.

$$\begin{array}{ccc} E_0 & \xrightarrow{\phi} & E \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \psi'} \\ E_1 & \xrightarrow{\phi'} & E_2 \end{array}$$

At a high level, in the SIDH proof of knowledge, the prover generates a random isogeny $\psi : E_0 \to E_1$, then computes its pushforward $\psi' : E \to E_2$ and the pushforward $\phi' : E_1 \to E_2$ of ϕ through ψ. The commitment is (E_1, E_2). The

challenge ch is sampled at random from $\{-1, 0, 1\}$. The response is ψ, ϕ' or ψ' if $\mathsf{ch} = -1$, $\mathsf{ch} = 0$ or $\mathsf{ch} = 1$ respectively. Upon receiving the response, the verifier checks that its domain and codomain are as expected with respect to the challenge. Moreover, when the challenge is 0, the verifier also checks that the degree of the response is d. Note that when proving soundness, one uses a more general version of $\mathcal{R}_{isog-d}$. We refer to [**?**] or [30] for further details. The SIDH proof of knowledge we have just sketched does not work for our relation $\mathcal{R}_d$ as the codomain curve of the witness isogeny is not public. We design a new proof of knowledge for the relation $\mathcal{R}_d$, prove its security, and discuss how it can be instantiated in practice.

3.1 The sigma protocol

Our proof of knowledge for $\mathcal{R}_d$ relies on the following diagram where $\psi_1 : E_1 \to E_2$ is the pushforward of ϕ through ψ_0.

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow{\scriptstyle \psi_0} & & \\ E_1 & \xrightarrow{\psi_1} & E_2 \end{array} \tag{1}$$

In the commitment phase, the prover generates a random isogeny $\psi_0 : E \to E_1$ of smooth degree d_{com} coprime with d. The commitment is the curve E_1. The challenge ch is sampled at random from $\{0, 1\}$. If $\mathsf{ch} = 0$, the prover returns the isogeny $\psi_0 : E \to E_1$. If $\mathsf{ch} = 1$, the prover computes the pushforward $\psi_1 : E_1 \to E_2$ of ϕ through ψ_0, and returns it as the response. In the verification phase, when $\mathsf{ch} = 0$, the verifier checks that the response ψ_0 is an isogeny from E to E_1 of degree d_{com}; and when $\mathsf{ch} = 1$, the verifier checks that the response ψ_1 is an isogeny of domain E_1 and of degree d. This sigma protocol, which we will denote by $\Sigma_{\mathcal{R}_d}$, is summarised in Figure 3. We prove that $\Sigma_{\mathcal{R}_d}$ is 2-special sound and honest verifier zero-knowledge.

Proposition 17. *Given two valid $\Sigma_{\mathcal{R}_d}$ transcripts for the same statement $\mathsf{stm} = E$ sharing the same commitment E_1 but with different challenges, one can efficiently extract an isogeny $\phi : E \to E'$ of degree d.*

Proof. Let $(E_1, 0, \psi_0 : E \to E_1)$ be the transcript with challenge 0, and let $(E_1, 1, \psi_1 : E_1 \to E_2)$ be the transcript with challenge 1. Since both transcripts are valid, then $\psi_0 : E \to E_1$ is an isogeny of degree d_{com} and $\psi_1 : E_1 \to E_2$ is an isogeny of degree d. Since d_{com} and d are coprime, then the pullback isogeny $\phi = \psi_0^* \psi_1$ of ψ_1 through ψ_0 is an isogeny of domain E and of degree d. □

Proving the zero-knowledge property is less straightforward. In fact, simulating the isogeny ψ_0 is easy, but in general, simulating ψ_1 without the knowledge of a witness ϕ is not, especially when d is not a smooth integer. At a high level, to simulate ψ_1, we first pick an elliptic curve E_0 of known endomorphism ring, say $E_0 : y^2 = x^3 + x$. We compute an isogeny $\psi_0' : E_0 \to E_1'$ of degree d_{com} and

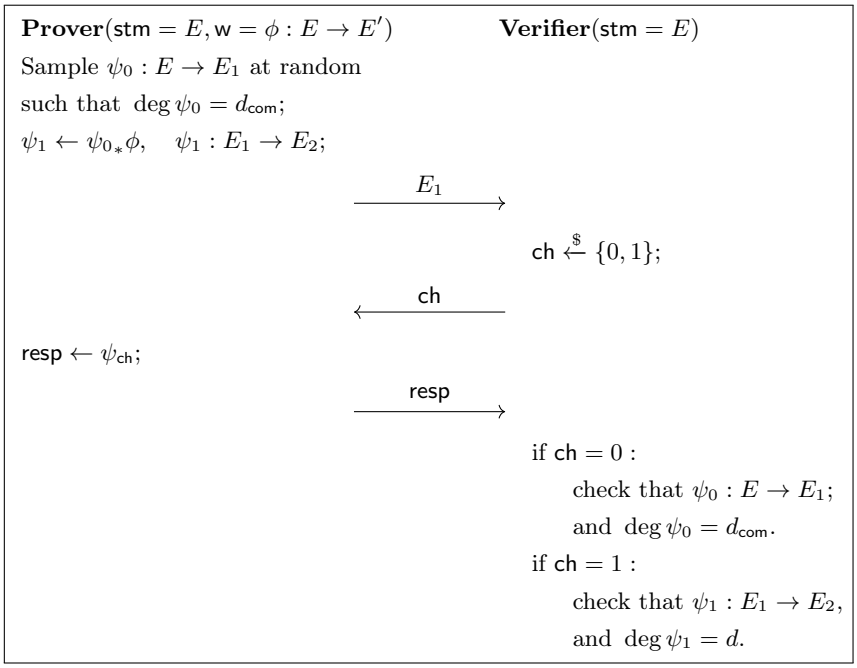


Fig. 3. $\Sigma_{\mathcal{R}_d}$: Proof of knowledge for $\mathcal{R}_d$

recover the endomorphism ring $\mathrm{End}(E_1')$ of E_1'. We then generate a uniformly random isogeny $\psi_1' : E_1' \to E_2'$ of degree d. Note that we generate the curve E_1' by computing an isogeny from a curve with known endomorphism ring because this will give us the capability of sampling random isogenies of degree d when d is not a smooth integer. Yet, after sampling ψ_1', we still need to ensure that its distribution is statistically indistinguishable from that of ψ_1 in a real transcript. This can be seen as a two level requirement as we need to ensure that:

1. the statistical distance between the simulated E_1' and the real E_1 is negligible, after which we ensure that
2. the statistical distance between the simulated (E_1', ψ_1') and the real (E_1, ψ_1) is negligible.

Let us assume that $d_{\mathsf{com}} = \ell^n$ is a power of a prime ℓ which does not divide d. Given a supersingular elliptic curve F and a positive integer T, let $\mathrm{Isog}_F(T)$ bet the set of all degree T isogenies of domain F. We consider the following distributions:

$$\mathcal{D}_{11}(k) = \{E_1' \mid \psi_0' \xleftarrow{\$} \mathrm{Isog}_E(\ell^k),\ E_1' = \mathrm{Im}(\psi_0')\},$$
$$\mathcal{D}_{12}(k) = \{E_1' \mid \psi_0' \xleftarrow{\$} \mathrm{Isog}_{E_0}(\ell^k),\ E_1' = \mathrm{Im}(\psi_0')\}.$$

For the first requirement, we want to ensure that $\mathcal{D}_{11}(k)$ and $\mathcal{D}_{12}(k)$ are statistically indistinguishable. This is achieved by the following Lemma.

Lemma 18. *Let λ be the security parameter. With the notations introduced above, the distributions $\mathcal{D}_{11}(k)$ and $\mathcal{D}_{12}(k)$ are statistically indistinguishable whenever $\log_\ell(p \cdot 2^{2\lambda}) \leq k$.*

Proof. This lemma is an application of Proposition 29 of [20] which states that when $\log_\ell(p \cdot 2^{2\lambda}) \leq k$, both $\mathcal{D}_{11}(k)$ and $\mathcal{D}_{12}(k)$ are statistically indistinguishable from the uniform distribution in the supersingular isogeny graph. □

For the second requirement, we define the following distributions:

$$\mathcal{D}_{21}(k) = \{(E_1', \psi_1') \mid \psi_0' \xleftarrow{\$} \mathrm{Isog}_E(\ell^k),\ E_1' = \mathrm{Im}(\psi_0'),\ \psi_1' = \psi_{0*}'\phi\},$$
$$\mathcal{D}_{22}(k) = \{(E_1', \psi_1') \mid \psi_0' \xleftarrow{\$} \mathrm{Isog}_E(\ell^k),\ E_1' = \mathrm{Im}(\psi_0'),\ \psi_1' \xleftarrow{\$} \mathrm{Isog}_{E_1'}(d)\}.$$

We want to ensure that the statistical distance between $\mathcal{D}_{21}(k)$ and $\mathcal{D}_{22}(k)$ is negligible. This brings us to the mixing time of non-backtracking walks in the supersingular ℓ-isogeny graph with level d Borel structure, which was studied in [**?**]. Given the field characteristic p, the degree d as in the relation $\mathcal{R}_d$, the prime ℓ that does not divide d and some integers k and m, we define

$$\tau(p, d, \ell, k) = \frac{1}{4}(p-1)^{1/2} \cdot \left(1 + \sqrt{d} \prod_{\substack{q \mid d, \\ q \text{ prime}}} (1 + \frac{1}{q})^{1/2}\right) \cdot \left(k + \frac{\ell - 1}{\ell + 1}\right) \cdot \ell^{-k/2}$$

and

$$n(p, d, \ell, m) = \min\{k \in \mathbb{Z} \mid \tau(p, d, \ell, k) \leq 2^{-m}\}.$$

Lemma 19. *Let λ be the security parameter. With the notations introduced above, the distributions $\mathcal{D}_{21}(k)$ and $\mathcal{D}_{22}(k)$ are statistically indistinguishable whenever $n(p, d, \ell, \lambda) \leq k$.*

Proof. This lemma is a direct application of Lemma 14 of [**?**] which states that the statistical distance between $\mathcal{D}_{21}(k)$ and $\mathcal{D}_{22}(k)$ is bounded by $\tau(p, d, \ell, k)$. □

We are now ready to prove that $\Sigma_{\mathcal{R}_d}$ is zero-knowledge.

Proposition 20. *Consider the notations introduced earlier in this section. Let $p > 2^{2\lambda}$ be the field characteristic, where λ is the security parameter. If $d_{\mathsf{com}} = \ell^n$ where n is such that $\max\{\log_\ell(p \cdot 2^{2\lambda}),\ n(p, d, \ell, \lambda)\} \leq n$, then there exists an efficient simulator that outputs random transcripts which are statistically indistinguishable from honest transcripts in $\Sigma_{\mathcal{R}_d}$.*

Proof. Given the statement $\mathsf{stm} = E$ and a challenge $\mathsf{ch} \in \{0, 1\}$, the simulator proceeds as follows. If $\mathsf{ch} = 0$, it generates a uniformly random isogeny $\psi_0' : E \to E_1'$, and returns $(E_1', 0, \psi_0')$ as the transcript. Since the isogeny ψ_0' is generated exactly like in the protocol, then the transcript $(E_1', 0, \psi_0')$ is valid and has the same distribution as transcripts with $\mathsf{ch} = 0$.

If $\mathsf{ch} = 1$, it picks a supersingular elliptic curve E_0 with known endomorphism ring $\mathrm{End}(E_0)$, say $E_0 : y^2 = x^3 + x$. It generates a uniformly random isogeny $\psi_0' : E_0 \to E_1'$ of degree d_{com}. It recovers the endomorphism ring $\mathrm{End}(E_1)$ of E_1. It then generates a uniformly random isogeny $\psi_1' : E_1' \to E_2'$ of degree d. The simulator then returns $(E_1', 1, \psi_1')$ as the simulated transcript. Since $\log_\ell(p \cdot 2^{2\lambda}) \leq n$, then by Lemma 18, the curve E_1' sampled by our simulator from the distribution $\mathcal{D}_{12}(n)$ can be assumed to come from the distribution $\mathcal{D}_{11}(n)$. Let us then assume that E_1' comes from the distribution $\mathcal{D}_{11}(n)$. Since $n(p, d, \ell, \lambda) \leq n$, then by Lemma 19, the distribution of (E_1', ψ_1') where ψ_1' is sampled uniformly at random from $\mathrm{Isog}_{E_1'}(d)$ is statistically indistinguishable from that of (E_1', ψ_1') where ψ_1' is obtained as the pushforward of ϕ through the isogeny ψ_0' connecting E to E_1'. Hence the simulated transcript $(E_1', 1, \psi_1')$ is statistically indistinguishable from a real transcript $(E_1, 1, \psi_1)$. □

Computational zero-knowledge. As shown in Proposition 20, aiming at statistical zero-knowledge implies taking d_{com} to be at least $p \cdot 2^{2\lambda}$. As we will see in the following paragraphs, this comes with a huge efficiency overhead. Here we consider computational zero-knowledge, which while relying on Assumption 21, allows to choose $d_{\mathsf{com}} \approx 2^{2\lambda}$ and hence obtain a faster sigma protocol.

Assumption 21 *Let $\phi : E \to E'$ be an isogeny of degree d, let E_0 be any supersingular elliptic curve with known endomorphism ring, and let $d_{\mathsf{com}} \geq 2^{2\lambda}$ be a smooth integer coprime to d. The following distributions are computationally indistinguishable:*

$$\mathcal{D}_1 = \{(E_1, \psi_1) \mid \psi_0 \xleftarrow{\$} \mathrm{Isog}_E(d_{\mathsf{com}}),\ E_1 = \mathrm{Im}(\psi_0),\ \psi_1 = \psi_{0*}\phi\},$$
$$\mathcal{D}_2 = \{(E_1', \psi_1') \mid \psi_0' \xleftarrow{\$} \mathrm{Isog}_{E_0}(d_{\mathsf{com}}),\ E_1' = \mathrm{Im}(\psi_0'),\ \psi_1' \xleftarrow{\$} \mathrm{Isog}_{E_1'}(d)\}.$$

Proposition 22. *Under Assumption 21, if $d_{\mathsf{com}} \geq 2^{2\lambda}$, then there exists an efficient simulator that outputs random transcripts which are computationally indistinguishable from honest transcripts in $\Sigma_{\mathcal{R}_d}$.*

Proof. The simulator proceeds exactly as in the proof of Proposition 20. This time, when the challenge is 1, we rely on Assumption 21 to claim that the transcript is computationally indistinguishable from a real transcript. □

Note that the only practical difference between $\Sigma_{\mathcal{R}_d}$ with computational zero-knowledge and $\Sigma_{\mathcal{R}_d}$ with statistical zero-knowledge is the degree of the commitment isogeny ψ_0. Hence this will be implicit in the parameters.

3.2 Practical considerations

In practice, one would like to use a PRISM friendly prime characteristic p which is as small as possible, that is $p = 2^a \cdot f - 1$ where f is a small cofactor and $a \approx 2\lambda$. In the previous section, no assumption was made on d. Nevertheless, the relation $\mathcal{R}_d$ is cryptographically interesting only when d is a non smooth integer, as it

becomes a hard relation. In the context of PRISM, we have $d = q(2^u - q) \approx p^2$ where q is a $u \approx a$ bits prime number. This implies that the witness ϕ is provided by its higher dimensional representation, which consists of $\phi(E[2^a])$.

The most convenient choice for d_{com} is to have it as a power of 2, since the 2^a torsion is available. For the computational zero-knowledge variant, we can have $d_{\mathsf{com}} = 2^a$, which means that the kernel of $\psi_0 : E \to E_1$ is rational. This implies that sampling the ψ_0 is straightforward, as one samples a generator of its kernel at random and computes the corresponding isogeny from its kernel. For the statistical zero-knowledge variant, we have that d_{com} is at least 2^{2a}, which means that the kernel of ψ_0 is not rational. One would need to split ψ_0 into a composition of isogenies of degree 2^a and consecutively sample such isogenies. In this case ψ_0 can be represented as a sequence of kernels.

Things become more delicate when one needs to compute the pushforward ψ_1 of ϕ through ψ_0. Since ϕ is provided by its higher dimensional representation, then the most common way to obtain ψ_1 is to exploit the equation $[d_{\mathsf{com}}]\psi_1 = \psi_0' \circ \phi \circ \widehat{\psi_0}$ where ψ_0' is the pushforward of ψ_0 through ϕ, which allows to evaluate ψ_1 on points of order coprime to d_{com}, and hence obtain its higher dimensional representation. The problem here is that having $d_{\mathsf{com}} = 2^a$ implies that one cannot use the equation $[d_{\mathsf{com}}]\psi_1 = \psi_0' \circ \phi \circ \widehat{\psi_0}$ to evaluate $\psi_1(E_1[2^a])$. A possible solution to this would be to set d_{com} to be odd. We need $p + 1$ to almost be a power of 2 so that ideal to isogeny algorithms (Qlapoti) are as efficient as possible. This does not allow to fit large smooth numbers into $p + 1$. One could try to use the smooth part of $p - 1$ as well in d_{com}, but the fact that $p + 1$ is almost a power of two implies that we have very little control on the smooth part of $p - 1$, which means that this smooth part is very small in practice. We hence narrow to the following two possibilities.

1. **Use powersmooth degree isogenies.** Here, as in [29] or in AprèsSQI [17], $d_{\mathsf{com}} = (\prod_{i=1}^{n} d_i^{e_i})^r$ is odd and powersmooth. The main advantage of this option is that we can obtain $\psi_1(E_1[2^a])$ as

$$\psi_1(E_1[2^a]) = [d_{\mathsf{com}}^{-1} \mod 2^a]\psi_0' \circ \phi \circ \widehat{\psi_0}(E_1[2^a])$$

   while maintaining the size of the base prime p as small as possible, that is, $p \approx 2^{2\lambda}$. There are two disadvantages. (i) One would need to iteratively work over extensions fields $\mathbb{F}_{p^{2k_i}}$ where $E[d_i^{e_i}] \subset E(\mathbb{F}_{p^{2k_i}})$, to compute $d_i^{e_i}$-isogenies and push points of order $d_j^{e_j}$ through them. (ii) When computing ψ_1, one would need to iteratively push ϕ through all the r $d_i^{e_i}$-isogenies for all $i \in \{1, \cdots, n\}$, which leads to ϕ being pushed nr times. A minor disadvantage is that since d_{com} is somewhat highly composite, the fact d needs to be coprime with d_{com} puts some restrictions on d, especially when one wants to fix the parameters once and run the protocol with different isogeny degrees d.
2. **Double the size of the prime** p**.** Here, one sets $p = 2^{4\lambda} \cdot f - 1$, and $d_{\mathsf{com}} = 2^{2\lambda}$ for the computational zero-knowledge variant. Assuming that ϕ is represented with $\phi(E[2^{2\lambda}])$, which is the case in PRISM for example, then

we can evaluate ψ_1 in one go as

$$\psi_1(E_1[2^{2\lambda}]) = [2^{2\lambda}]\psi_1(E_1[2^{4\lambda}]) = \psi_0' \circ \phi \circ \widehat{\psi_0}(E_1[2^{4\lambda}]).$$

For the statistical zero-knowledge variant where $d_{\mathsf{com}} = 2^{2r\lambda}$, one sample ψ_0 as a composition of r isogenies of degree $2^{2\lambda}$ and evaluate ψ_1 by performing r pushforwards of ϕ. The main advantage is that the scheme is more efficient. In fact, in the computational zero-knowledge version for example, the degrees of the isogenies are unchanged, hence the efficiency overhead only comes from the field arithmetic, which is expected to be 2 to 4 times slower as the field size doubles. Nevertheless, since one works over $\mathbb{F}_{p^2}$ and only one pushforward of ϕ is computed, this overhead is smaller compared to when working over field extensions and computing several pushforwards. A minor advantage is that the scheme is easier to describe. The main disadvantage is that we will over with a larger field, which implies that the resulting signature will be larger.

In the following section, we go for the second option and work with a prime $p = 2^{4\lambda} \cdot f - 1$. In Appendix A, we describe the more compact but less efficient variant where the kernel of d_{com} is not rational.

4 SPIDER

In this section, we specify the inner workings of $\Sigma_{\mathcal{R}_d}$ and turn it into a digital signature which we call SPIDER[4].

4.1 $\Sigma_{\mathcal{R}_d}$: specification

We provide the public parameters, the algorithmic details of the commitment, response and verification phases. Moreover, we also specify how to sample uniformly at random from $\mathcal{R}_d$, which corresponds to key generation when seeing $\Sigma_{\mathcal{R}_d}$ as an identification scheme.

Public parameters. For a security level λ, we use a prime $p = 2^{2a} \cdot f - 1$ where $a \approx 2\lambda$ and f is a small integer such that p is a prime. Let $E_0 : y^2 = x^3 + x$. Since $p \equiv 3 \mod 4$, then E_0 is supersingular and its endomorphism ring $\mathrm{End}(E_0) \simeq \mathcal{O}_0$ is known. Let $d_{\mathsf{com}} = 2^{ar}$, with $r = 1$ for the computational zero-knowledge variant, or r is such that

$$\max\{\log(p \cdot 2^{2\lambda}),\ n(p, d, 2, \lambda)\} \leq ar.$$

We assume that d is such that for any isogeny $\phi : E \to E'$ of degree d, $\phi(E[2^a])$ is an efficient representation of ϕ. We hence obtain the public parameters

$$\mathsf{pp}_{\Sigma_{\mathcal{R}_d}} = (p, a, d, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0).$$

[4] SPIDER: **S**upersingular **P**rime **I**sogeny **DE**g**R**ee. Note that *prime degree isogeny* is the correct expression, but we are sure that the reader prefers SPIDER to SPDERI as we do.

Algorithm 1 $\mathcal{R}_d$.Sample

Require: $\mathsf{pp}_{\Sigma_{\mathcal{R}_d}} = (p, a, d, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0)$
Ensure: $(E, \phi) \in \mathcal{R}_d$
1: Sample a random prime $N > p^2 > p \cdot 2^{2\lambda}$
2: Sample a uniformly random left ideal I_0 of $\mathcal{O}_0$ of norm N
3: Translate I_0 into an isogeny $\phi_0 : E_0 \to E$ using Qlapoti
4: Sample a uniformly random left ideal J of $\mathcal{O}_0$ of norm d
5: Translate $J \cap I_0$ into its corresponding isogeny $\phi \circ \phi_0 : E_0 \to E'$ using Qlapoti
6: Retrieve ϕ from ϕ_0 and $\phi \circ \phi_0$.
7: **return** (E, ϕ)

Algorithm 2 $\Sigma_{\mathcal{R}_d}$.Commit

Require: $\mathsf{pp}_{\Sigma_{\mathcal{R}_d}}$, $(E, \phi) \in \mathcal{R}_d$
Ensure: $\mathsf{com}_0 \in \mathbb{Z}_{2^a}^r$, $\mathsf{com}_1 = E_1$
1: Sample $(s_u)_u \xleftarrow{\$} \mathbb{Z}_{2^a}^r$
2: Set $E_{00} = E$
3: Let $Q_1 \in E_{00}[2^a]$ be a canonically generated point of order 2^a
4: **for** $u \in [1, \cdots, r]$ **do**
5: Canonically complete P_u to a basis $\langle P_u, Q_u \rangle = E_{0(u-1)}[2^a]$
6: Set $R_u = P_u + [s_u]Q_u$
7: Compute $\psi_{0u} : E_{0(u-1)} \to E_{0u}$ of kernel $\langle R_u \rangle$
8: Set $Q_{u+1} = \psi_{0u}(Q_u)$
9: $\mathsf{com}_0 = (s_u)_u$, $\mathsf{com}_1 = E_{0r}$.
10: **return** com_0, com_1

Sampling from $\mathcal{R}_d$. If we followed Section 3.1 very well, then we may already have an idea on how to sample from $\mathcal{R}_d$. In fact, one proceeds in a similar way as the simulator when the challenge is 1. One generates a uniformly random isogeny $\phi_0 : E_0 \to E$ of large degree $N > p \cdot 2^{2\lambda}$ and one recovers the endomorphism ring $\mathrm{End}(E_1)$ of E_1. One then generates a uniformly random isogeny $\phi : E \to E'$ of degree d, to obtain $(E, \phi) \in \mathcal{R}_d$. As proved in Section 3.1, the distribution of such (E, ϕ) is statistically indistinguishable from the uniform distribution on $\mathcal{R}_d$. In practice, to sample the isogeny ϕ_0, one first samples its left ideal I_0 of $\mathcal{O}_0$, then one translates I_0 into an isogeny using Qlapoti. Since this isogeny will be used later in the protocol, either when computing pushforwards or evaluating other isogenies on points, it is best to choose its degree $N > p \cdot 2^{2\lambda}$ to be a prime number, so that it is always coprime with degrees of other isogenies and to the order of the torsion points used in the protocol. To generate ϕ, one samples a random left ideal J of norm d, computes the ideal $I_0 \cap J$ which corresponds to an isogeny $\phi \circ \phi_0 : E_0 \to E'$ where $\phi : E \to E'$ is the d-isogeny corresponding to the ideal $I_{0*}J$. One translates $I_0 \cap J$ into the isogeny $\phi \circ \phi_0$ using Qlapoti, then one retrieves ϕ from ϕ_0 and $\phi \circ \phi_0$. This process is summarised in Algorithm 1.

Algorithm 3 $\Sigma_{\mathcal{R}_d}$.Response

Require: pp_{Σ_d}, (E, ϕ), com_0, ch
Ensure: resp: a representation of an isogeny
if $\mathsf{ch} = 0$ **then**
  return com_0
Parse com_0 as $(s_u)_u \in \mathbb{Z}_{2^a}^r$
Compute $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$ from $(s_u)_u$ as in Algorithm 2
Parse $\phi : E \to E'$ as $\phi(E[2^{2a}])$
Set $E_{00} = E$, $\phi_0 = \phi$, $E_{0u} = \mathrm{Im}(\psi_{0u})$
for $u \in [1, \cdots, r]$ **do**
  Recover $\phi_{u-1}(\ker \psi_{0u})$ from $\phi_{u-1}(E_{0(u-1)}[2^{2a}])$
  Compute $\psi'_{0u} = \phi_{u-1*}\psi_{0u}$ whose kernel is $\phi_{u-1}(\ker \psi_{0u})$
  Compute $\phi_u(E_{0u}[2^a]) = \psi'_{0u} \circ \phi_{u-1} \circ \widehat{\psi_{0u}}(E_{0u}[2^{2a}])$
  if $u < r$ **then**
    Recover $\phi_u(E_{0u}[2^{2a}])$ from $\phi_u(E_{0u}[2^a])$
$\mathsf{resp} = \phi_r(E_{0r}[2^a])$.
return resp

Commitment. The commitment consists of sampling a random isogeny $\psi_0 : E \to E_1$ of degree 2^{ar}. We repeatedly sample r non-backtracking isogenies of degree 2^a, to obtain ψ_0 as a composition $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$. To sample ψ_{01}, we sample its kernel from $E[2^a]$ be generating a canonical basis (P_1, Q_1) of $E[2^a]$, sampling a random integer $s \in \mathbb{Z}_{2^a}$ and setting $\ker \psi_{01} = P_1 + [s]Q_1$. One then computes the corresponding isogeny ψ_{01} together with $\psi_{01}(Q_1)$. Note that by design, $\langle Q_1 \rangle \cap \ker \psi_{01} = 0$. The point $\psi_{01}(Q_1)$ is used when sampling the next isogeny ψ_{02}, as we need to ensure that $\langle \psi_{01}(Q_1) \rangle \cap \ker \psi_{02} = 0$ so that there is no backtracking. One then repeats this process r times to sample ψ_0 which can be represented using $(s_1, \cdots, s_r) \in \mathbb{Z}_{2^a}^r$. The first part $\mathsf{com}_0 = (s_1, \cdots, s_r) \in \mathbb{Z}_{2^a}^r$ of the commitment is kept secret, while the second part $\mathsf{com}_1 = E_1$ is sent to the verifier. This process is summarised in Algorithm 2. One should note that since the 2^{2a} torsion is available, in the statistical zero-knowledge variant where $r > 1$, the first factors of ψ_0 can be chosen to have degree 2^{2a} and only the last factor will have degree 2^a when r is odd.

Response. If $\mathsf{ch} = 0$, the prover returns $(s_1, \cdots, s_r) \in \mathbb{Z}_{2^a}^r$, which is a representation of the isogeny $\psi_0 : E \to E_1$ of degree d_{com}. If $\mathsf{ch} = 1$, the prover needs to compute the pushforward $\psi_1 : E_1 \to E_2$ of ϕ through ψ_0 which they will return as the response. During commitment, the prover computed ψ_0 as $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$. They will now iteratively push ϕ through each ψ_{0i}. For $\psi_{01} : E \to E_{01}$, they evaluate ϕ on $\ker \psi_{01}$ and compute the pushforward $\psi'_{01} : E' \to E'_{01}$ of ψ_{01} through ϕ. They then obtain the evaluation of the pushforward $\phi_1 : E_{01} \to E_{01}$ of ϕ through ψ_{01} as

$$\phi_1(E_{01}[2^a]) = \psi'_0 \circ \phi \circ \widehat{\psi_0}(E_{01}[2^{2a}]).$$

Algorithm 4 $\Sigma_{\mathcal{R}_d}$.Verify

Require: $\mathsf{pp}_{\Sigma_{\mathcal{R}_d}}$, com_1, ch, resp
Ensure: accepted/rejected
Parse com_1 as E_1
if $\mathsf{ch} = 0$ **then**
  Parse resp as $(s_u)_u \in \mathbb{Z}_{2^a}^r$
  if $(s_u)_u$ defines a cyclic isogeny $E \to E_1$ of degree 2^{ar} **then**
    return accepted
  return rejected
Parse resp as $\psi_1(E_1[2^a])$.
if $\psi_1(E_1[2^a])$ defines an isogeny $E_1 \to E_2$ of degree d **then**
  return accepted
return rejected

From the efficient representation $\phi_1(E_{01}[2^a])$ of ϕ_1, they recover $\phi_1(E_{01}[2^{2a}])$ with which they will compute the pushforward of ϕ_1 through ψ_{02}. they repeat this process r times. On the last round they do not need to recover $\phi_r(E_{01}[2^{2a}])$ from $\phi_r(E_{0r}[2^a])$, as the data $\phi_r(E_{0r}[2^a]) = \psi_1(E_1[2^a])$ is a representation of ψ_1. Note that this representation is in general[5] made of the image of a deterministic basis of $E_1[2^a]$. This response mechanism is summarised in Algorithm 3.

Verification When $\mathsf{ch} = 0$, then one parses the response as $(s_1, \cdots, s_r) \in \mathbb{Z}_{2^a}^r$ and com_1 as E_1, and checks that it defines a cyclic isogeny $\psi_0 : E \to E_1$ of degree 2^{ar} computed as described in the commitment algorithm. When $\mathsf{ch} = 1$, one parses the response as $\psi_1(E_1[2^a])$ and com_1 as E_1, and checks that it is the higher dimensional representation of an isogeny $E_1 \to E_2$ of degree d. This verification mechanism is summarised in Algorithm 4.

4.2 The SPIDER signature scheme

The soundness error of $\Sigma_{\mathcal{R}_d}$ is $1/2$. To bring it down to negligible, we need to do parallel repetitions. Once this is done, we then apply the Fiat-Shamir transform to obtain a digital signature which is EUF-CMA secure. First, we discuss some well known techniques which we use to optimise the resulting signature scheme.

Using challenges with fixed weight. From the discussion in Section 3.2, it is clear that the workload of the prover and the size of the transcript when the challenge is 1 are very high/big compared to when the challenge is 0. It is hence desirable to limit the number of times the prover gets the challenge 1 in the parallel repetitions. Furthermore, fixing the number of 1's would also allow to have a signer runtime which does not directly depend on the challenge. This is known

[5] This assumes that the verifier has access to E_1. When it is not the case, for example when deriving a signature scheme from the sigma protocol, then one needs to include E_1 as well.

Algorithm 5 SPIDER.KGen

Require: $\overline{\mathsf{pp}}$
Ensure: $(\mathsf{pk}, \mathsf{sk}) \in \mathcal{R}_d$
1: Get $\mathsf{pp}_\Sigma = (p, a, d, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0)$ from $\overline{\mathsf{pp}}$
2: $(E, \phi) \leftarrow \mathcal{R}_d.\mathsf{Sample}(\mathsf{pp}_\Sigma)$
3: Set $\mathsf{pk} = E$, $\mathsf{sk} = \phi$
4: **return** $(\mathsf{pk}, \mathsf{sk})$

in the literature as fixed weight challenges. It consists of the following: when doing the t parallel repetitions to bring down the soundness error to negligible, one samples the challenge $\mathsf{ch} = (\mathsf{ch}_1, \cdots, \mathsf{ch}_t) \in \{0,1\}^t$ for the parallel repetitions uniformly at random from the set $\mathcal{C}_w(t) := \{\mathsf{ch} \in \{0,1\}^t \mid \mathsf{wt}(\mathsf{ch}) = w\}$ where $\mathsf{wt}(\mathsf{ch})$ is the hamming weight of ch and w is a fixed integer between 1 and t. The soundness error of the resulting sigma protocol is $\binom{t}{w}^{-1}$. For a given security level λ, we need to ensure that t and w are chosen in such a way that $\binom{t}{w}^{-1} \leq 2^{-\lambda}$.

Commitment recoverability. In $\Sigma_{\mathcal{R}_d}$, the commitment $\mathsf{com}_1 = E_1$ can be recovered from the challenge ch and the response $\mathsf{resp} = (s_u)_u$ when $\mathsf{ch} = 0$. This means that when the challenge is 0, the commitment can be omitted from the transcript $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$, as it can be obtained from $(\mathsf{ch}, \mathsf{resp})$. In the case where the challenge is 1, then the data used to represent $\psi_1 : E_1 \to E_2$ will also include the commitment E_1. At the end of the verification, the verifier checks that the challenge was honestly generated using the recovered commitments. This allows to reduce the size of the signature when applying the Fiat-Shamir transform.

Public parameters and key generation. For a security level λ, the public parameters of SPIDER include those of $\Sigma_{\mathcal{R}_d}$, to which one adds the number t of repetitions, the weight w of the challenges, and a hash function

$$H : \{0,1\}^* \to \mathcal{C}_w(t) := \{\mathsf{ch} \in \{0,1\}^t \mid \mathsf{wt}(\mathsf{ch}) = w\};$$

where $\binom{t}{w}^{-1} \leq 2^{-\lambda}$ and H is modelled as a random oracle. We hence obtain

$$\overline{\mathsf{pp}} = (p, a, d, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H).$$

The key generation in SPIDER consists of sampling at random from $\mathcal{R}_d$ which is achieved using Algorithm 1. This is summarised in Algorithm 5.

Signing. To sign a message msg, one generates t commitments $(\mathsf{com}_0^{(z)}, \mathsf{com}_1^{(z)})_z$ and sets $\overline{\mathsf{com}}_0 = (\mathsf{com}_0^{(z)})_z$, $\overline{\mathsf{com}}_1 = (\mathsf{com}_1^{(z)})_z$. The challenge is then obtained as $\overline{\mathsf{ch}} = (\mathsf{ch}_z)_z \leftarrow H(\mathsf{pk} || \overline{\mathsf{com}}_1 || \mathsf{msg})$. For $z \in \{1, \cdots, t\}$, one recovers $\mathsf{resp}_z = \Sigma_{\mathcal{R}_d}.\mathsf{Response}((\mathsf{pk}, \mathsf{sk}), \mathsf{com}_0^{(z)}, \mathsf{ch}_z)$. One should note that in $\Sigma_{\mathcal{R}_d}$, one assumes that the verifier knows the commitment E_1 before the response phase. This implies that when the challenge is 1, the representation of $\psi_1 : E_1 \to E_2$ does

Algorithm 6 SPIDER.Sign

Require: $\overline{\mathsf{pp}}$, $(\mathsf{pk}, \mathsf{sk})$, a message msg
Ensure: A signature σ
1: Get t, H and $\mathsf{pp}_\Sigma = (p, a, d, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0)$ from $\overline{\mathsf{pp}}$
2: **for** $z \in [1, \cdots, t]$ **do**
3: $\quad \mathsf{com}_0^{(z)}, \mathsf{com}_1^{(z)} \leftarrow \Sigma_{\mathcal{R}_d}.\mathsf{commit}(\mathsf{pp}_\Sigma, (\mathsf{pk}, \mathsf{sk}))$
4: Set $\overline{\mathsf{com}}_0 = (\mathsf{com}_0^{(z)})_z$, $\overline{\mathsf{com}}_1 = (\mathsf{com}_1^{(z)})_z$
5: $\overline{\mathsf{ch}} = (\mathsf{ch}_z)_z \leftarrow H(\mathsf{pk}||\overline{\mathsf{com}}_1||\mathsf{msg})$
6: **for** $z \in [1, \cdots, t]$ **do**
7: $\quad$ **if** $\mathsf{ch}_z = 0$ **then**
8: $\quad\quad \mathsf{resp}_z = \Sigma_{\mathcal{R}_d}.\mathsf{Response}(\mathsf{pp}_\Sigma, (\mathsf{pk}, \mathsf{sk}), \mathsf{com}_0^{(z)}, \mathsf{ch}_z)$
9: $\quad$ **else**
10: $\quad\quad \mathsf{resp}_z = (\mathsf{com}_1^{(z)}, \Sigma_{\mathcal{R}_d}.\mathsf{Response}(\mathsf{pp}_\Sigma, (\mathsf{pk}, \mathsf{sk}), \mathsf{com}_0^{(z)}, \mathsf{ch}_z))$
11: Set $\overline{\mathsf{resp}} = (\mathsf{resp}_z)_z$
12: **return** $\sigma = (\overline{\mathsf{ch}}, \overline{\mathsf{resp}})$

not includes E_1. Here, since the verifier does not yet have access to E_1 nor can recover it himself, then its also included in resp_z. We make this explicit the pseudocode (Algorithm 6) summarising the signature process in SPIDER. Let $\overline{\mathsf{resp}} = (\mathsf{resp}_z)_z$. The signature is $\sigma = (\overline{\mathsf{ch}}, \overline{\mathsf{resp}})$.

Verification. Let $\sigma = (\overline{\mathsf{ch}}, \overline{\mathsf{resp}})$ be a signature for a message msg under a key pair $(\mathsf{pk}, \mathsf{sk})$. For the 0 challenges ch_z in $\overline{\mathsf{ch}}$, one parses resp_z as $(s_u)_u$ and computes the corresponding non-backtracking isogeny $E \to E_1^{(z)} = \mathsf{com}_1^{(z)}$. For the 1 challenges ch_z in $\overline{\mathsf{ch}}$, one parses resp_z as $\mathsf{com}_1^{(z)} = E_1^{(z)}$ and a representation of $\psi_1^{(z)} : E_1^{(z)} \to E_2^{(z)}$ and checks that $\psi_1^{(z)}$ is an isogeny of degree d. Lastly, one checks that $\overline{\mathsf{ch}} = H(\mathsf{pk}||\overline{\mathsf{com}}_1||\mathsf{msg})$ where $\overline{\mathsf{com}}_1 = (\mathsf{com}_1^{(z)})_z$. This process is summarised in Algorithm 7.

Security. As highlighted earlier, SPIDER is obtained by applying the Fiat-Shamir transform to the sigma protocol $\Sigma_{\mathcal{R}_d}$. Since $\Sigma_{\mathcal{R}_d}$ is sound and honest verifier zero-knowledge, then SPIDER is EUF-CMA secure under the assumption that computing isogenies of degree d is hard.

Problem 23. Given a uniformly random supersingular elliptic curve E and a random non smooth integer d, compute an isogeny $\varphi : E \to E'$ of degree d.

For Problem 23 to be hard, d needs to be a non smooth integer. As in PRISM, we will assume that d is of the form $d = q(2^a - q)$ where q is a prime having $a \approx 2\lambda$ bits. This implies that Problem 23 is the same as Problem 15 but without access to the SPEDIO oracle.

Proposition 24. *Assume that the hash function H is modelled as a random oracle and $d = q(2^a - q)$ where q is a random prime having $a \approx 2\lambda$ bits. If $d_{\mathsf{com}} =$*

Algorithm 7 SPIDER.Verify

Require: $\overline{\mathsf{pp}}$, pk, msg, σ
Ensure: accepted/rejected
Get t, H and $\mathsf{pp}_\Sigma = (p, a, d, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0)$ from $\overline{\mathsf{pp}}$
Parse σ as $\overline{\mathsf{ch}} = (\mathsf{ch}_z)_z$ and $\overline{\mathsf{resp}} = (\mathsf{resp}_z)_z$
for $z \in [1, \cdots, t]$ **do**
    if $\mathsf{ch} = 0$ **then**
        Parse resp_z as $(s_u)_u \in \mathbb{Z}_{2^a}^r$
        Compute the cyclic 2^{ar}-isogeny $E \to E_1^{(z)}$ defined by $(s_u)_u$
        Set $\mathsf{com}_1^{(z)} = E_1^{(z)}$
    else
        Parse resp_z as $\mathsf{com}_1^{(z)} = E_1^{(z)}$ and $\psi_1^{(z)}(E_1^{(z)}[2^a])$
        $\mathsf{bool} \leftarrow \Sigma_{\mathcal{R}_d}.\mathsf{Verify}(\mathsf{pp}_\Sigma, \mathsf{com}_1^{(z)}, \mathsf{ch}, \psi_1^{(z)}(E_1^{(z)}[2^a]))$
        if $\mathsf{bool} = \mathsf{rejected}$ **then**
            return rejected
Set $\overline{\mathsf{com}}_1 = (\mathsf{com}_1^{(z)})_z$
$\overline{\mathsf{ch}}' \leftarrow H(\mathsf{pk}||\overline{\mathsf{com}}_1||\mathsf{msg})$
if $\overline{\mathsf{ch}} = \overline{\mathsf{ch}}'$ **then**
    return accepted
return rejected

ℓ^n where n is such that $\max\{\log_\ell(p \cdot 2^{2\lambda}),\ n(p, d, \ell, \lambda)\} \leq n$, (resp. $d_{\mathsf{com}} > 2^{2\lambda}$), then SPIDER is EUF-CMA secure under the assumption that Problem 23 is hard (resp. Problem 23 is hard and Assumption 21 holds).

5 An identity-based signature from SPIDER and PRISM

We describe our identity-based signature. At a high level, the master public key is a uniformly random supersingular elliptic curve $\mathsf{mpk} = E$ and the master secret key is an isogeny $\mathsf{msk} = (\phi_0 : E_0 \to E)$. Given a user id id, the secret of the corresponding user is an isogeny $\mathsf{usk}_{\mathsf{id}} = (\phi_{\mathsf{id}} : E \to E')$ of degree $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$ where $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(E||\mathsf{id})$ and $\mathsf{H}_{\mathsf{primes}_a}$ is the hash function used in PRISM. This means that the Master generates $\mathsf{usk}_{\mathsf{id}}$ by running the PRISM signature on the message $\mathsf{msg} = \mathsf{id}$ with the PRISM key pair $(\mathsf{pk}, \mathsf{sk}) = (\mathsf{mpk} = E, \mathsf{msk} = \phi_0)$. To sign a message msg, the owner of the id id runs the SPIDER signature scheme on the message msg with the SPIDER key pair $(\mathsf{pk}, \mathsf{sk}) = (\mathsf{mpk} = E, \mathsf{usk}_{\mathsf{id}} = \phi_{\mathsf{id}})$.

Remark 25. There exists several frameworks [33] that allow to generically design Identity-Based Signatures. While the design of SPIDER-IBS looks like it could fit into that of Bellare, Namprempre and Neven [5], it actually doesn't. In fact, this framework requires a trapdoor samplable relation (see [5, Sec. 4.1]) which in our setting would translate to one being able to sample a tuple $(d, \phi : E \to E')$ where $d = q(2^a - q)$ with q being a large prime and $\deg \phi = d$, when given a random supersingular elliptic curve E. The latter is hard and it is up to some extent related to the security of PRISM.

Algorithm 8 SPIDER-IBS.MKgen

Require: $\mathsf{pp} = (p, a, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H, \mathsf{H}_{\mathsf{primes}_a})$
Ensure: $(\mathsf{mpk}, \mathsf{msk})$
1: Sample a random prime $N_0 > p^2$
2: Sample a uniformly random left ideal I_0 of $\mathcal{O}_0$ of norm N_0
3: Translate I_0 into an isogeny $\phi_0 : E_0 \to E$ using Qlapoti
4: Set $\mathsf{msk} = \phi_0$, $\mathsf{mpk} = E$
5: **return** $(\mathsf{mpk}, \mathsf{msk})$

Algorithm 9 SPIDER-IBS.KeyDer

Require: pp, $\mathsf{mpk} = E$, $\mathsf{msk} = \phi_0$, id
Ensure: $\mathsf{usk}_{\mathsf{id}} = \phi_{\mathsf{id}}$
1: **return** PRISM.Sign$(\mathsf{mpk}, \mathsf{msk}, \mathsf{id})$

5.1 SPIDER-IBS: specification

The public parameters are made of those of SPIDER without the degree d, to which one adds the hash function $\mathsf{H}_{\mathsf{primes}_a}$ used in PRISM. That is

$$\mathsf{pp} = (p, a, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H, \mathsf{H}_{\mathsf{primes}_a}).$$

The master key generation is similar to that of PRISM, except that the base prime $p = 2^{2a} \cdot f - 1$ with $a \approx 2\lambda$ is 2x larger. The master key generation is summarised in Algorithm 8. The user derivation consists of generating a PRISM signature on the message $\mathsf{msg} = \mathsf{id}$ with the master key pair $(\mathsf{mpk}, \mathsf{msk}) = (E, \phi_0)$. Note that the prime p is 2x larger but the degree of the isogeny $\phi_{\mathsf{id}} : E \to E'_{\mathsf{id}}$ is $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$ where $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(\mathsf{mpk}||\mathsf{id})$. This time, one returns $\phi_{\mathsf{id}}(E[2^{2a}])$ as the signature, instead of $\phi_{\mathsf{id}}(E[2^a])$. This choice is not strictly necessary but it saves the user who owns id from evaluating of ϕ_{id} on $E[2^{2a}]$ when generating a SPIDER signature. The user key derivation is summarised in Algorithm 8.

To sign a message msg, the owner of id generates a SPIDER signature on msg. The verifier will then verify that the signature is a valid SPIDER signature under SPIDER public parameters where the degree d is $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$ with $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(\mathsf{mpk}||\mathsf{id})$. The signing and verification algorithms in SPIDER-IBS are summarised in Algorithm 10 and Algorithm 11 respectively.

5.2 Security arguments

We now discuss the security arguments for SPIDER-IBS. It would have been great to obtain the security of SPIDER-IBS directly from that of PRISM and SPIDER, but there is a small subtlety which prevents us from doing so. This is somewhat related to the fact that we do not have a samplable relation, as highlighted in Remark 25. In fact, in the $\mathsf{Exp}_\lambda^{\mathrm{IBS}}()$ game, corrupting users is equivalent to having access to the SPEDIO oracle in PRISM. Hence, when attempting to forge signatures in the IBS setting, the adversary is in fact trying

Algorithm 10 SPIDER-IBS.Sign

Require: pp, $\mathsf{mpk} = E$, id, $\mathsf{usk}_{\mathsf{id}} = \phi_{\mathsf{id}}$, msg
Ensure: A signature σ
1: Compute $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(E||\mathsf{id})$
2: Set $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$
3: Get $p, a, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H$ from pp
4: Set $\overline{\mathsf{pp}}_{\mathsf{id}} = (p, a, d_{\mathsf{id}}, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H)$
5: **return** SPIDER.$\mathsf{Sign}(\overline{\mathsf{pp}}_{\mathsf{id}}, (\mathsf{mpk}, \mathsf{usk}_{\mathsf{id}}), \mathsf{msg})$

Algorithm 11 SPIDER-IBS.Verify

Require: pp, $\mathsf{mpk} = E$, id, msg, σ
Ensure: accepted/rejected
1: Compute $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(E||\mathsf{id})$
2: Set $d_{\mathsf{id}} = q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$
3: Get $p, a, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H$ from pp
4: Set $\overline{\mathsf{pp}}_{\mathsf{id}} = (p, a, d_{\mathsf{id}}, r, t, w, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0, H)$
5: **return** SPIDER.$\mathsf{Verify}(\overline{\mathsf{pp}}_{\mathsf{id}}, \mathsf{mpk}, \mathsf{msg}, \sigma)$

to forge a SPIDER signature but with access to the SPEDIO oracle. Since in PRISM we assume that Problem 15 is hard, then the access to SPEDIO does not give them any advantage in computing large prime degree isogenies (apart from those they obtained from the oracle). We further conjecture that it does not give them any advantage in proving the knowledge of any such isogeny. We formalise this in the following assumption.

Assumption 26 *The SPIDER signature for relations $\mathcal{R}_d$ where $d = q(2^a - q)$ with $a \approx 2\lambda$ and $q \in \mathsf{primes_a}$ remains EUF-CMA secure when the adversary in the EUF-CMA game is further equipped*[6] *with the SPEDIO oracle which returns isogenies of degree $d' = q'(2^a - q')$ where $q' \in \mathsf{primes_a}$.*

Theorem 27. *If Assumption 26 holds, then SPIDER-IBS is UF-CID-CMA secure.*

Proof. Let $\mathcal{A}$ be an adversary who plays the $\mathsf{Exp}_{\lambda}^{\mathrm{IBS}}()$ game. Let E be the master public key generated by the challenger. Then we can replace the $O_{\mathsf{corr}}()$ oracle in the game by the SPEDIO oracle. In fact, a call to $O_{\mathsf{corr}}()$ produces an isogeny $\mathsf{usk} : E \to E'$ of degree $q_{\mathsf{id}}(2^a - q_{\mathsf{id}})$ where $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(E||\mathsf{id})$. The replacement proceeds as follows: given the identity id, one computes the $q_{\mathsf{id}} = \mathsf{H}_{\mathsf{primes}_a}(E||\mathsf{id})$ and calls SPEDIO on E and id to obtain $\mathsf{usk} : E \to E'$. Since the access to $O_{\mathsf{corr}}()$ is the same as the access to SPEDIO, then an adversary who produces a valid forgery $(\mathsf{msg}^*, \mathsf{id}^*, \sigma^*)$ is identical to an adversary who forges SPIDER signatures when equipped with the SPEDIO oracle. By Assumption 26, its advantage is negligible.

[6] Forgeries with respect to degrees queried to the oracle are rejected.

5.3 Instantiation

For the security parameters 128, 192 and 256, we can use the highly conservative prime $2^{512} \cdot 3^5 - 1, 2^{771} \cdot 19 - 1$ and $2^{1025} \cdot 61 - 1$ respectively

SPIDER: We have $(\mathsf{pk}, \mathsf{sk}) = (E, \phi)$ where the isogeny ϕ can be represented by $(E', \phi(P), \phi(Q))$, (P, Q) being a deterministic basis of $E[2^a]$. As argued in Section 3.2, we need the evaluation of ϕ on a 2^{2a}-torsion basis of E to compute the pushforward. We will then consider (P, Q) to be a basis of $E[2^{2a}]$ instead. Any elliptic curve $C/\mathbb{F}_{p^2}$ can be represented in Montgomery model by a coefficient $A \in \mathbb{F}_{p^2}$ such that $C : y^2 = x^3 + Ax^2 + x$. This representation requires $2\log_2(p) \approx 8\lambda$ bits. A point $R \in C(\mathbb{F}_q)$ is usually represented by its x-coordinate, which is an element $x \in \mathbb{F}_q$. In our setting, we have $\phi(P), \phi(Q) \in E'[2^{2a}]$. Any point $R \in E'[2^{2a}]$ can be represented by its coordinate $(u, v) \in \mathbb{Z}/2^{2a}\mathbb{Z} \times \mathbb{Z}/2^{2a}\mathbb{Z}$ in a deterministic basis (P', Q') of $E'[2^{2a}]$. This representation requires $4a$ bits. Since $\deg(\phi)$ is known, $(\phi(P), \phi(Q))$ can be represented by three integers u, v, u', such that the coordinates of $(\phi(P)$ and $\phi(Q))$ are respectively $(u, v), (u', v')$. In this way, the fourth integer v' can be recover using Weil pairing and discrete logarithm in base 2^{2a}. Finally, one needs $2\log_2(p) \approx 8\lambda$ bits for pk and $2\log_2(p) + 6a \approx 20\lambda$ for sk.

For the signature, we have $\sigma = (\mathsf{ch}, \mathsf{resp})$ where $\mathsf{ch} \in \mathcal{C}_w \subset \{0,1\}^t$ and $\mathsf{resp} = (\mathsf{resp}_1, \ldots, \mathsf{resp}_t)$ with $\mathsf{resp}_i \in \{\psi_0^i, \psi_1^i\}$. Any isogeny ψ_0^i has degree 2^{ar} and can be represented by r elements of $\mathbb{Z}/2^a\mathbb{Z}$. This representation requires $a \cdot r$ bits. Any isogeny ψ_1^i has a HD representation of the form $(E_1, E_2, \psi_1^i(P_1), \psi_1^i(Q_1))$ where E_1 and E_2 are respectively the domain and codomain curve, and P_1, Q_1 is a deterministic basis of $E_1[2^a]$. This representation requires $4\log_2(p) + 3a \approx 22\lambda$ bits. Finally, one needs $t + (t - \omega)a \cdot r + \omega(4\log_2(p) + 3a) \approx t + 2r\lambda(t - \omega) + 22\lambda\omega$ bits. As discussed in Section 4.2, t and ω are such that $\binom{t}{w}^{-1} \leq 2^{-\lambda}$. For the security parameters 128, 192 and 256, the middle values of (t, ω) are respectively $(210, 33), (324, 48)$ and $(450, 62)$, giving the signature size in Table 2 for the computational zero-knowledge variant.

SPIDER-IBS: The master public key mpk is an elliptic curve. Then it can be represented with $2\log_2(p) \approx 8\lambda$. The user secret key usk has the same form as sk in SPIDER. We can represent it with 20λ bits. The signature has the same size as in SPIDER.

λ	$\lvert\mathsf{mpk}\rvert$ (bytes)	$\lvert\mathsf{usk}\rvert$ (bytes)	$\lvert\mathsf{sig}\rvert$ (kilobytes)
128	130	226	17.03
192	194	338	37.92
256	258	450	58.18

Table 2. Master public key, user secret key and signature sizes in SPIDER-IBS.

5.4 Comparison with existing isogeny-based IBS

Keys and signature sizes. The following table gives the comparison of our IBS with that in [14] and [48].

Scheme	$\lvert\mathsf{mpk}\rvert$	$\lvert\mathsf{usk}\rvert$	$\lvert\mathsf{sig}\rvert$
[48]	$S_0\lceil\log_2 p\rceil$	$T_1S_1(\lceil\log_2 S_0\rceil + \lceil\log_2 N\rceil)$	$T_1S_1\lceil\log_2 p\rceil +$ $T_1T_2(\lceil\log_2 S_1\rceil + \lceil\log_2 N\rceil)$
[14]	$S_0\lceil\log_2 p\rceil$	$T_1S_1(\lceil\log_2 p\rceil + \lceil\log_2 N\rceil)$	$T_1S_1\lceil\log_2 p\rceil +$ $T_1T_2(\lceil\log_2 S_1\rceil + \lceil\log_2 N\rceil)$
SPIDER-IBS	$2\lceil\log_2 p\rceil$	$2\lceil\log_2 p\rceil + 12\lambda$	$t + 2(t-\omega)\lambda \cdot r +$ $\omega(4\lceil\log_2(p)\rceil + 6\lambda)$

Table 3. Comparison of isogeny-based IBS

The IBS schemes in [48] and [14] use supersingular curves defined over $\mathbb{F}_p$. The class group structure is known and its order is $N \approx \sqrt{p}$. The parameters (S_0, S_1) and (T_1, T_2) are related to the (lossy) identification scheme. These protocols all use isogeny group actions which are subject to quantum sub-exponential attacks, implying that the primes needed to instantiate them have several thousand bits. For example, as recommended in [13], a prime of 3000 to 4000 bits needs to be used for isogeny group action schemes claiming 128 bits of security. By the description in [14], $\{0, \ldots, S_0\}^{T_1S_1}$ and $\{0, \ldots, S_1\}^{T_1T_2}$ are challenge spaces used by the master and the user respectively, where $S_i = 2^{\eta_i} - 1$, $T_1 < S_0$ and $T_2 < S_1$ for some positive integers η_i $(i = 0, 1)$. One needs $(S_0+1)^{T_1S_1} > 2^\lambda$ and $(S_1+1)^{T_1T_2} > 2^\lambda$, which means $2^{\eta_0 T_1 S_1} > 2^\lambda$ and $2^{\eta_1 T_1 T_2} > 2^\lambda$ i.e, $T_1S_1 > \lambda/\eta_0$ and $T_1T_2 > \lambda/\eta_1$. These constraints seem to optimise the size of the signature sig in detriment of the size of the master public key and user secret key. For example, for a prime p such that $\log_2 p \approx 3000$ and with $(\eta_0, \eta_1, T_1) = (5, 5, 1)$, we obtain (T_1S_1, T_1T_2) is $(31, 26)$, the master public mpk has size 1875 bytes, the user secret key has size 7000 bytes and the signature has size 18 kilobytes. While this signature size is only 1.06x larger compared to that of SPIDER-IBS, the master public and the user secret key are 14.42x and 30.97x larger.

Efficiency. SPIDER is an order of magnitude more efficient compared to existing isogeny-based identity-based signatures [48,14]. In fact, as highlighted above, they all use isogeny group actions which are subject to quantum sub-exponential attacks, and their secure instantiation needs to use primes of 3000 bits or more, while we use a prime of size 500 bits. Moreover, since they class group structure (used in those constructions) is not known for those large primes, one needs to rely on higher dimensional isogenies such as qt-PEGASIS [19] or its predecessors [18,40,39] when evaluating isogeny group actions. One also needs to use a generalisation of CSI-FiSh [28, Section 2.4] to larger parameters, which is slower in general as it uses qt-PEGASIS as a subroutine.

For example, considering the parameters $\log_2 p \approx 3000$ and $(\eta_0, \eta_1, T_1) = (5, 5, 1)$ (such that $(T_1 S_1, T_1 T_2) = (31, 26)$) for the IBS in [14], we get that the master key generation, the user key generation, the signature and the verification cost $2 + 2S_0 = 64$, $2T_1 S_1 = 62$, $2T_1 T_2 = 52$ and $2T_1 T_2 = 52$ group action evaluations respectively. Each group action evaluation costs about $\log_2 p$ (2,2,2,2) isogenies in characteristic p when using qt-PEGASIS [19]. Hence, the master key generation, the user key generation, the signature and the verification cost about $64{\cdot}3000 = 192\,000$, $62{\cdot}3000 = 186\,000$, $52{\cdot}3000 = 156\,000$ and $52{\cdot}3000 = 156\,000$ (2,2,2,2) isogenies in characteristic p with $\log_2 p \approx 3000$.

While for SPIDER-IBS, we use a prime p such that $\log_2 p \approx 512$. The master key generation and user key generation basically cost one call to ideal to isogeny, which computes about 512 $(2, 2)$ isogenies over $\mathbb{F}_{p^2}$. The signing consists of computing $t+w = 243$ isogenies of degree $2^{2\lambda}$ while verification consists of computing $t - w = 177$ isogenies of degree $2^{2\lambda}$ and $w = 33$ dimension 2 isogenies of degree $2^{2\lambda}$, that is $33 \cdot 256 = 8448$ (2,2) isogenies isogenies over $\mathbb{F}_{p^2}$.

Follow up work. While the numbers above show that SPIDER-IBS is several orders of magnitude faster compared to the IBS schemes in [48,14], a concrete implementation of those schemes and SPIDER-IBS will allow a more precise estimate of the efficiency gap between them. We leave this for future work.

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A SPIDER 2: working over extension fields

In this section, we specify the inner workings of $\Sigma_{\mathcal{R}_d}$ when we use an odd commitment degree $d_{\mathsf{com}} = d_c^r$ where $\prod_{i=1}^{n} d_i^{e_i}$ and $E[d_i^{e_i}]$ is accessible over a small extension field. This variant allows the use of a base p which is roughly $2^{2\lambda}$, resulting in a more compact scheme. We specify the public parameters, the commitment, the response and the verification algorithms. Note that the sampling from $\mathcal{R}_d$ mechanism does not change as it is independent of the prime factorisation of the degree of the commitment isogeny.

Algorithm 12 $\Sigma_{\mathcal{R}_d}$.Commit

Require: pp, $(E, \phi) \in \mathcal{R}_d$
Ensure: $\mathsf{com}_0 \in \mathbb{Z}_{d_c}^r$, $\mathsf{com}_1 = E_1$
Get $(d_i)_i, (e_i)_i, r$ from pp
Sample $(s_u)_u \xleftarrow{\$} \mathbb{Z}_{d_c}^r$
Set $F_0 = E$
For $i = 1, \cdots, n$, let $Q_{1i} \in E[d_i^{e_i}]$ be a canonically generated point of order $d_i^{e_i}$
for $u \in [1, \cdots, r]$ **do**
    For $i = 1, \cdots, n$, canonically complete Q_{ui} to a basis $\langle P_{ui}, Q_{ui}\rangle = F_{u-1}[d_i^{e_i}]$
    Set $R_{ui} = P_{ui} + [s_u]Q_{ui}$, for $i = 1, \cdots, n$
    Set $G_{u0} = F_{u-1}$, $\phi_{u0} = \mathrm{Id}_{G_{u0}}$
    for $i \in [1, \cdots, n]$ **do**
        Compute $\phi_{ui} : G_{u(i-1)} \to G_{ui}$ of kernel $\phi_{u(i-1)} \circ \cdots \circ \phi_{u0}(R_{ui})$
        For all $v \in [i+1, \cdots, n]$, compute $\phi_{ui}(\phi_{u(i-1)} \circ \cdots \circ \phi_{u0}(R_{uv}))$
        For all $j \in [1, \cdots, n]$, compute $\phi_{ui}(\phi_{u(i-1)} \circ \cdots \circ \phi_{u0}(Q_{uj}))$
    For all $i \in [1, \cdots, n]$, set $Q_{(u+1)i} = \phi_{un} \circ \cdots \circ \phi_{u1}(Q_{uj}))$
    $F_u = G_{un}$
$\mathsf{com}_0 = (s_u)_u$, $\mathsf{com}_1 = F_r$.
return com_0, com_1

Public parameters. For a security level λ, we use a prime $p = 2^a \cdot f - 1$ where $a \approx 2\lambda$ and f is a small integer such that p is a prime. Let $E_0 : y^2 = x^3 + x$. Since $p \equiv 3 \mod 4$, then E_0 is supersingular and its endomorphism ring $\mathrm{End}(E_0) \simeq \mathcal{O}_0$ is known. Let $d_c = \prod_{i=1}^n d_i^{e_i} \approx 2^{\lambda/2}$ coprime with d such that $E_0[d_i^{e_i}]$ is available over a very small extension field $\mathbb{F}_{p^{2k_i}}$. Let r be the integer such that $d_{\mathsf{com}} = d_c^r$. For the computational zero-knowledge variant, $r = 4$. For the statistical zero-knowledge variant, r is such that

$$\max\{\log_{\ell_c}(p \cdot 2^{2\lambda}),\ n(p, d, \ell_c, \lambda)\} \leq \log_{\ell_c}(d_{\mathsf{com}}) = \log_{\ell_c}(d_c^r),$$

where ℓ_c is the largest prime below the bound $\prod_{i=1}^n d_i^{e_i - 1}(d_i + 1)$. Since the only difference between the computational zero-knowledge variant and the statistical zero-knowledge variant is the degree d_{com} of the commitment isogeny ψ_0, the value of r in the parameters indicates in which case we are. This is because for computational zero-knowledge we have $d_{\mathsf{com}} = d_c^r \approx 2^{2\lambda}$ while for statistical zero-knowledge $d_{\mathsf{com}} = d_c^r > p \cdot 2^{2\lambda} > 2^{4\lambda}$. We hence obtain the public parameters which are

$$\mathsf{pp} = (p, a, d, (d_i)_i, (e_i)_i, (k_i)_i, r, E_0, \mathrm{End}(E_0) \simeq \mathcal{O}_0).$$

Commitment. The commitment consists of sampling a random isogeny $\psi_0 : E \to E_1$ of degree d_c^r. Since only the d_c-torsion is accessible, then we need to repeatedly sample r non-backtracking isogenies of degree d_c, to obtain ψ_0 as a composition $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$. To sample ψ_{01}, we sample its kernel from $E[d_c]$ be generating a canonical basis (P_1, Q_1) of $E[d_c]$, sampling a random integer $s \in \mathbb{Z}_{d_c}$ and setting $\ker \psi_{01} = P_1 + [s]Q_1$. One then computes the corresponding

Algorithm 13 $\Sigma_{\mathcal{R}_d}$.Response

Require: pp, (E, ϕ), com_0, ch
Ensure: resp: a representation of an isogeny
1: **if** $\mathsf{ch} = 0$ **then**
2: **return** com_0
3: Get $(d_i)_i, (e_i)_i, r$ from pp
4: Set $d_c = \prod_i d_i^{e_i}$
5: Parse com_0 as $(s_u)_u \in \mathbb{Z}_{d_c}^r$
6: Compute $\psi_0 : E \to E_1$ as $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$ where $\ker \psi_{0u}$ is defined by s_u
7: Parse $\phi : E \to E'$ as $\phi(E[2^a])$
8: Set $F_0 = E$, $\phi_0 = \phi$, $F_0' = E'$, $F_u = \mathrm{Im}(\psi_{0u})$
9: **for** $u \in [1, \cdots, r]$ **do** ▷ This loop is similar to the one in step 5 of Algorithm 12
10: Evaluation ϕ_{u-1} on $\ker \psi_{0u}$
11: Compute $\psi_{0u}' = \phi_{u-1*}\psi_{0u}$ whose kernel is $\phi_{u-1}(\ker \psi_{0u})$
12: Compute $\phi_u(F_u[2^a]) = [d_c^{-1} \mod 2^a]\psi_{0u}' \circ \phi_{u-1} \circ \widehat{\psi_{0u}}$
13: $\mathsf{resp} = \phi_r(F_r[2^a])$.
14: **return** resp

isogeny ψ_{01} together with $\psi_{01}(Q_1)$. Note that by design, $\langle Q_1 \rangle \cap \ker \psi_{01} = 0$. The point $\psi_{01}(Q_1)$ is used when sampling the next isogeny ψ_{02}, as we need to ensure that $\langle \psi_{01}(Q_1) \rangle \cap \ker \psi_{02} = 0$ so that there is no backtracking. One then repeats this process r times to sample ψ_0 which can be represented using $(s_1, \cdots, s_r) \in \mathbb{Z}_{d_c}^r$. Sampling ψ_{01} as described above requires accessing $E[d_c] \subset E(\mathbb{F}_{p^{2k}})$ where $k = \mathrm{LCM}((k_i)_i)$ which is relatively larger than the k_is in practice. To avoid working with points defined over $\mathbb{F}_{p^{2k}}$, we write $E[d_c] = \sum_{i=1}^n E[d_i^{e_i}]$, work with each $E[d_i^{e_i}]$ separately and compute ψ_{01} as a composition $\psi_{01} = \phi_{1n} \circ \cdots \circ \phi_{11}$ of isogenies ϕ_{1i} of respective degree $d_i^{e_i}$. The first part $\mathsf{com}_0 = (s_1, \cdots, s_r) \in \mathbb{Z}_{d_c}^r$ of the commitment is kept secret, while the second part $\mathsf{com}_1 = E_1$ is sent to the verifier. This process is summarised in Algorithm 12.

Response. If $\mathsf{ch} = 0$, the prover returns $(s_1, \cdots, s_r) \in \mathbb{Z}_{d_c}^r$, which is a representation of the isogeny $\psi_0 : E \to E_1$. If $\mathsf{ch} = 1$, the prover needs to compute the pushforward $\psi_1 : E_1 \to E_2$ of ϕ through ψ_0 which they will return as the response. During commitment, the prover has computed ψ_0 as $\psi_0 = \psi_{0r} \circ \cdots \circ \psi_{01}$. They will now iteratively push ϕ through each ψ_{0i}. For ψ_{01}, it evaluates ϕ on $\ker \psi_{01}$ and computes the pushforward ψ_{01}' if ψ_{01} through ϕ. It then obtains the higher dimensional representation of the pushforward ϕ_1 of ϕ through ψ_{01} as $\phi_1(E_{01}[2^a]) = [d_c^{-1} \mod 2^a]\psi_{01}' \circ \phi \circ \widehat{\psi_{01}}$. Repeat this process r times. This response mechanism is summarised in Algorithm 13.

Verification When $\mathsf{ch} = 0$, then one parses the response as $(s_1, \cdots, s_r) \in \mathbb{Z}_{d_c}^r$ and com_1 as E_1, and checks that it defines an isogeny $\psi_0 : E \to E_1$ of degree d_c^r computed as described in the commitment algorithm. When $\mathsf{ch} = 1$, one parses the response as $\psi_1(E_1[2^a])$ and com_1 as E_1, and checks that it is the higher

dimensional representation of an isogeny $E_1 \to E_2$ of degree d. This process is summarised in Algorithm 14.

Algorithm 14 $\Sigma_{\mathcal{R}_d}$.Verify

Require: pp, com_1, ch, resp
Ensure: accepted/rejected
Get $(d_i)_i, (e_i)_i, r$ from pp
Parse com_1 as E_1
if $\mathsf{ch} = 0$ **then**
    Parse resp as $(s_u)_u \in \mathbb{Z}_{d_c}^r$
    if $(s_u)_u$ defines an isogeny $E \to E_1$ of degree d_c^r **then**
        return accepted
    return rejected
Parse resp as $\psi_1(E_1[2^a])$.
if $\psi_1(E_1[2^a])$ defines an isogeny $E_1 \to E_2$ of degree d **then**
    return accepted
return rejected